

W6 *FOR THE STARS WITHOUT NUMBER ROLE-PLAYING GAME*

DISTANT LIGHTS

BY KEVIN CRAWFORD



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CREATING BORDERWORLD OUTPOSTS FOR YOUR CAMPAIGN

BUILD BORDERWORLD OUTPOSTS

DISTANT LIGHTS GIVES THE BUSY GM THE TOOLS THEY NEED TO FASHION REMOTE PLANETARY OUTPOSTS, SECLUDED DEEP-SPACE ASTEROID HABITATS, OR BRASH COLONIAL FOOTHOLDS ON FAR-FLUNG WORLDS. TABLES AND GUIDES ARE PROVIDED FOR GENERATING OUTPOST ADVENTURE HOOKS, TROUBLED NPCS, AND RESTIVE NATIVE POPULATIONS. ADDITIONAL MATERIALS GIVE PLAYERS THE PRICES AND GUIDES THEY NEED FOR ESTABLISHING THEIR OWN INTERSTELLAR OUTPOSTS ON DISTANT WORLDS.

WHILE *DISTANT LIGHTS* WAS WRITTEN FOR THE *STARS WITHOUT NUMBER* ROLE-PLAYING GAME, THE TOOLS IN THIS BOOK ARE LARGELY SYSTEM-NEUTRAL, AND USABLE WITH YOUR OWN FAVORITE GAMING SYSTEM OR SCI-FI SETTING OF CHOICE.

DISTANT LIGHTS

BORDERWORLD OUTPOSTS FOR *STARS WITHOUT NUMBER*



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A FAR AND LONELY PLACE

Sci-fi tales are full of elaborate worlds and vast societies. One can hardly open a pulp space adventure without meeting planet-covering megacities or timelessly ancient cultures, and authors are often eager to tell you how many billions of inhabitants make up one star empire or another. "Big" is a standard adjective for the future.

Yet another vein of adventure is found in the remote border-worlds of a setting, the dusty and peripheral planets which have no human habitation whatsoever except for a bunker and a wind-scoured landing pad. These worlds have no cities or teeming millions, but only a relative handful of workers, overseers, or exiles to carry out their purposes. They are the wilderness-beset villages or hidden hamlets of the sci-fi genre.

And like those borderlands villages, they have excellent usefulness to adventurers. The basic necessities of life can often be had at such places, but they're far enough from any greater power that the PCs only need concern themselves with the goodwill of the outpost's leadership. The only laws that matter are those that can be effectively imposed by the locals, and those laws are very few. A reckless band of interstellar freebooters can get up to all kinds of trouble in such places without having to worry about invincible swarms of planetary security officers or the panopticon gaze of security cameras around every corner.

Distant Lights gives the GM the guidelines they need for making these remote outposts, helping a creator to build them in a plausible fashion and make sure that they pull their creative weight in adventure hooks and playable table content. With the tables and procedures in this book, a GM can quickly generate a serviceable interstellar colony or remote corporate outpost that can give the PCs something useful to do and someplace interesting to conduct their business.

The second part of the book includes rules for PCs who want to establish their own planetary outpost. While full-scale planetary colonization is covered in the *Suns of Gold* supplement, this book gives them the tools they need for building smaller, less costly outposts of civilization.

When To Use This Book

For GMs who aren't certain when to bother with creating an outpost, a basic rule can be kept in mind: **planetary outposts are space villages**. Whenever a fantasy-genre GM would need or use a remote village, a sci-fi GM can profit by having a planetary outpost.

Is there a planet where some human presence is necessary for an adventure or for the setting's logic, but you don't want to build out an entire human society? Put an outpost on it. Perhaps there really are a large number of native humans elsewhere on the planet, perhaps it's totally devoid of other intelligent life, but this one offworld landing spot is the most reasonable place for the PCs to touch down. If they want to talk to the locals, buy supplies, refuel their ship, or get local services, this outpost is where they can do those things and pick up on useful local adventure hooks. It's the space keep on the space borderlands.

Small settlements such as these tend to be much easier to handle than vast sci-fi cities, because all the really important NPCs can be summarized on a 3x5 card. You know who exactly is likely to be responding to PC actions or goals, and you also have a much better idea of their available resources.

With that in mind, it can be convenient to use the guidelines in this book to generate a quick, simple outpost with a generic patron and widely-applicable purpose and then tuck it in your GM folder for later use. Whenever you're in a situation where a world ought reasonably to have some human occupation but you don't have time to flesh it out, you can pull out this premade outpost, fill in a few pertinent blanks, and plop it down on the planet.

This book can also be handy when the PCs decide they need to infiltrate or assault an outpost, whether that place is a den of nefarious space cultists, a pirate haven, or the branch facility of a business rival. The defense facilities listed in the book can give a hint as to the kind of military force an outpost may be able to muster, and listings of important infrastructure facilities make good pick lists for busy saboteurs.





THE OUTPOST'S MOOD

The tables in this book are flexible and can produce a wide variety of different outposts. Sometimes it can be difficult to stitch the table results together into a cohesive whole, however, and so it can be useful to establish a basic mood to the outpost first, in order to have some guidance when reconciling the future elements of the place.

Essentially, whenever you have a set of random roll results that don't seem like they'd make sense, take a moment to think about how the mood of the place might reconcile them. If you make the overall mood responsible for the most obvious contradictions of the place, the mood itself will be more apparent to the PCs and have a bigger impact on play there.

The adjacent table offers twelve different possibilities for the overall mood. Most of them are tilted toward trouble, as conflict is most fruitful of adventuring content.

The table below gives a suggestion as to where you might make the mood most obvious. You can play it up in the situations listed below, giving it some obvious impact that the PCs will notice soon after they land. Once you set the tone, they'll be all the more likely to notice it elsewhere.

d6	Where is the Mood Most Obvious?
1	The outpost's relationship with its patron or the purpose for which the patron built it
2	The way the local workers and inhabitants behave when off duty
3	The way the outpost interacts with outside powers or intelligent native inhabitants
4	The current project or goal of the outpost's leadership
5	The last major event or crisis that the outpost faced or experienced
6	Some special service or opportunity present at the outpost hinges on it

What is the Outpost's Overall Mood?

- 1 Desperate.** Everything seems to be just barely hanging together and no one thinks it is going to improve soon.
- 2 Reckless.** They're doing things that an outside observer can see are very dangerous or imprudent, but they seem blind to the risk.
- 3 Despairing.** The locals are unhappy, but they're too hopeless to seek any improvement in their situation.
- 4 Excited.** Most of them are thrilled to be here and excited by the prospects at hand.
- 5 Bitter.** They don't want to be here and resent that they are stuck in this place.
- 6 Contented.** It's a peaceful or satisfying life here for most, comfortable and relaxed.
- 7 Ambitious.** People here can see a better life within reach, and they're willing to do reckless or unwise things to get it.
- 8 Secretive.** A lot of people here are doing things they don't want outsiders to know about, or perhaps even their own neighbors.
- 9 Nationalistic.** They're proud of their patron and determined to advance its profit or glory by doing the best job possible here.
- 10 Chaotic.** Things aren't working the way they're supposed to, whether due to local failings or unexpected realities about the planet.
- 11 Drudging.** Everything is about the outpost's role or purpose, and ordinary recreation or human pastimes are left to suffer for it.
- 12 Overconfident.** The outpost's is boldly moving forward with a plan or purpose that is premised on a belief that is deeply and catastrophically in error.

THE OUTPOST'S CONTEXT

When detailing the outpost, you need to decide why it was put here in the first place. While there's an element of world-building verisimilitude involved in this decision, the choice's more important role is to give you a basic background to fall back on when you need to come up with some detail or motivation related to the place. If the outpost was built for reason X, then whatever's going on there probably has to do with reason X too.

Who put it there? What planetary government, private corporation, religious cult, pirate band, criminal cabal, research institute, or ethnic exodus was behind the founding of the outpost? Was it the work of a single group, or is this place the fruit of an alliance, whether intended or circumstantial? The outpost may or may not still consider itself beholden to its patrons, but the founding population will likely have been drawn from their numbers.

Why was it put here? What is it about this planet or patch of space that makes it a good spot for an outpost? What does this location have that the founding group would have wanted? A minority group determined to hide from a cosmos seen as cruel and oppressive might find great value in a completely isolated zone of deep space, while a rapacious imperial hegemon might be throwing out forward bases for naval ships and spy rings halfway across the sector, regardless of the habitability of any particular site. Whatever the site, something about it is providing something the founding group wanted to get.

What do the locals think? If the world or area is inhabited by other sentients, what do they think about the outpost? A planet might have a population of much lower-tech natives, trapped by a lack of resources at a low tech level and eager to get the materials they need to bootstrap to more advanced industry. It might have several other outposts planted on it, each backed by a different rival power. Particularly large populations are unlikely to be monolithic in their views of the

outpost, and a continent split up into several dozen native nations might result in several dozen different diplomatic attitudes toward the outpost. Alien sentients on the planet may not even have a comprehensible attitude, as their opinions are based on motivations absent in human diplomacy.

What do the neighboring planets think? Establishing a naval base one drill away from a populated world is unlikely to be viewed kindly by that planet's government. Multiple worlds might lay claim to a particular planet, and this long-running dispute is why none of them have been able to plant more than this outpost on the world. A planet may be viewed as a strategic weak point by one world, a holy planet by another, a barbaric den of subhuman savages by a third, and be totally unknown to a fourth world that doesn't have the tech necessary to reach it. These outside players are likely to become factors in play when you need an outside antagonist or mission-giver to turn up at the outpost.

What's going to happen to it? It may be that the future of the outpost is irrelevant to your campaign; it's just a brief waystation for the PCs to land on before venturing onward. If you expect the outpost to be a recurring place in your campaign, however, you should take a minute to think about the likely arc of its future. Is it going to grow? Shrink? Collapse in a welter of blood and fire? Be seized by a more powerful neighbor? You don't have to plot out a specific timeline for it, but if you have a good idea of what lies in the colony's future, you can foreshadow it in various adventure hooks and slice-of-life scenes around the place. It could be that the actions of the PCs substantially change the outpost's future, or they may get attached enough to the NPCs there to actually care what becomes of them six months later. If the locals have made any friends among the PC party, it's quite possible that some job offers might be following should the outpost find itself in a future that has a use for heavily-armed freebooters.



EXTERNAL RELATIONS

All but the most hermetically sealed outposts will have some relations with extra-planar powers and visitors, even if they may only be agents of the outpost's founding patron. These tables can be used for inspiration when answering some of the important questions about the outpost's founding, purpose, and relationship with its patron.

These results should be read in the context of the tags you've established for the world. If the planet is an ocean world, an outpost established for mining is either processing mineral-rich seawater or conducting sea-floor excavation.

Note also that just because it was founded for a particular purpose, it doesn't mean that its current main product is necessarily the same. Situations change, and old infrastructure may need to serve new ends.

d6	How Do They Treat Adventurers?
1	They're cheerfully willing to deal with them.
2	They're wary, but will cut deals
3	They're useful, disposable, dangerous assets.
4	Indifferent, as long as there's no trouble.
5	They're viewed as dangerous space hobos.
6	Different factions have different attitudes.

d8	What Patron Founded It?
1	A neighboring planetary power
2	A despised or hated group
3	A far trader or corporation
4	A pirate, exiled noble, or renegade
5	An explorer or adventurer
6	A religious or philosophical sect
7	An educational or scientific institution
8	An alien species, perhaps now gone

d10	What Is Its Current Main Product?
1	Valuable scientific research on something
2	Profitable trade with natives or passing ships
3	Legitimacy for a claim to local rule
4	Valuable native goods or services
5	Precious loot or salvage from the past
6	Propaganda for a cause beloved by its patron
7	Native mercenaries, laborers, or slaves
8	Defense against a threat to the patron
9	Venting some societal pressure for a patron
10	Useful manufactures built by the outpost

d4	Is It Serving Its Purpose?
1	No. It is failing badly in its intended role.
2	Partially, or serving in an unexpected way
3	Yes, it's managing to do what was intended.
4	It's somehow doing better than expected.

d12	Why Was It Founded?
1	Refueling stop and supply depot
2	Black site for illegal or immoral deeds
3	Refuge for a persecuted or hated group
4	Mining or extracting a local resource
5	Religious retreat or pilgrimage site
6	Pleasure resort or hunting zone
7	Archaeological or looting forward base
8	Crash site or forced landing
9	Naval garrison or colonial military base
10	Embassy or cultural center to natives
11	Criminal or pirate base of operations
12	Prison colony or place of exile

d20	What Twist Exists In Its Context?
1	It's a re-founding of a former failed outpost.
2	It's been forgotten, lost, or written off.
3	The patron is exerting almost no control.
4	A new patron seized it from the founders.
5	The patron's expectations are changing.
6	The rationale for its founding was mistaken.
7	The patron has unreasonable demands.
8	The patron is getting false information.
9	The outpost's control is contested by a rival.
10	The outpost is widely misrepresented.
11	The outpost's real purpose is a secret one.
12	The ostensible patron is a false flag.
13	It's been greatly over-hyped to colonists.
14	A great hazard was kept quiet by its patron.
15	It's the private satrapy of a patron faction.
16	It's cooperating with a rival of the patron.
17	Its repurposing patron support for a goal.
18	It's somehow reliant on a different patron.
19	It presents a wholly false facade to the world.
20	It's highly resistant to a changing situation.

LOCAL RELATIONS

Just as the outpost needs to deal with the sector beyond its system, it also needs to handle relations with its neighbors on the planet. These tables assume that there are two main groups of concern: the outpost itself and the major native polity nearby. If there are multiple groups in the area, the tables can be used to sketch some relationships between them.

Very low-tech worlds rarely have the communications and force-projection infrastructure necessary to have a global government. Even those of TL3 or TL4 may well be broken up into numerous nations or regional factions. These different native groups will likely have different attitudes toward the outpost. Some may view it as the vanguard of an invasion, while others could see it as a precious trading depot, or even a weapon to use against their local rivals.

d6 What Outpost Dwellers Interact Most?

- | | |
|---|---|
| 1 | Interpreters, diplomats, or dragomans |
| 2 | Low-level workers or native commoners |
| 3 | Military officials or soldiers |
| 4 | Artists, scientists, or other culture-bearers |
| 5 | Priests or religiously-active persons |
| 6 | Supervisors, foremen, labor team leaders |

d8 What's The Worst the Outpost's Done?

- | | |
|---|---|
| 1 | Seized useful or religiously-important land |
| 2 | Violated a deeply-held taboo or law |
| 3 | Killed an important or well-loved person |
| 4 | Outraged the other with a cultural element |
| 5 | Stolen an artifact or an important resource |
| 6 | Deposed or humiliated an important leader |
| 7 | Driven people from their former homes |
| 8 | Catalyzed some local disaster or crisis |

d10 What Use is One Group to the Other?

- | | |
|----|---|
| 1 | A specific tech or knowledge they have. |
| 2 | Skilled or unskilled labor the other has. |
| 3 | A leader's prestige depends on their support. |
| 4 | They can intimidate or placate a rival. |
| 5 | Their culture fascinates the other group. |
| 6 | They have useful physical qualities. |
| 7 | They produce a vitally important resource. |
| 8 | They maintain a critical disaster preventive. |
| 9 | They're instrumental in achieving a goal. |
| 10 | Scapegoats for a failure by the other group. |

d4 How Frequent are Interactions?

- | | |
|---|--|
| 1 | They almost never engage with the other. |
| 2 | They have infrequent dealings. |
| 3 | They are often involved with each other. |
| 4 | It's a constant daily involvement. |

d12 What's the Current Problem?

- | | |
|----|--|
| 1 | Laws. Conflicting law codes contend. |
| 2 | Violence. People are being hurt or killed. |
| 3 | Love. Forbidden relationships cause tension. |
| 4 | Greed. Someone's grasping for more. |
| 5 | Hate. A group is abhorrent to the other. |
| 6 | Contempt. A group is considered inferior. |
| 7 | Ignorance. A group is critically mistaken. |
| 8 | Folly. A group's leadership is incompetent. |
| 9 | Naiveté. A group's too trusting or idealistic. |
| 10 | Faith. Religious are clashing dangerously. |
| 11 | Fear. A group is terrified of the other. |
| 12 | Recklessness. Consequences are ignored. |

d20 What's Happening Right Now?

- | | |
|----|---|
| 1 | A colonist and a native are in awkward love. |
| 2 | A major festival is coming up. |
| 3 | A group is stealing things for a good reason. |
| 4 | An unsolved murder is spiking tensions. |
| 5 | A custom from one group is getting popular. |
| 6 | A legal dispute is caught between two laws. |
| 7 | An allied group is under outside pressure. |
| 8 | One side is exploiting an agreement unfairly. |
| 9 | A group demands a problematic home site. |
| 10 | One side is trying to impose a cultural taboo. |
| 11 | A recent shared crisis is being met by both. |
| 12 | One side rescued a group from the other. |
| 13 | Isolationists are getting active after a clash. |
| 14 | A religious or cultural missionary is active. |
| 15 | A rich gift was offered from one side. |
| 16 | An invalid was cured by tech or native lore. |
| 17 | A dire natural omen has disturbed believers. |
| 18 | Usual trade has been corrupted by someone. |
| 19 | One side is temporarily reliant on the other. |
| 20 | A new group is trying to get involved. |

HISTORICAL OUTPOST EVENTS

Most outposts can be adequately sketched simply by deciding who founded it and why they did so. Other times, however, a GM might want to flesh out a place's history a little more thoroughly, if only to provide more adventure hooks for the PCs there.

The events described below are meant to be the sort that have current repercussions. A revolt, raid, or past disaster that had no effect that the PCs can't notice within an hour of landing is generally not worth detailing. If you're going to put some past into your outpost, make sure it's a past that still echoes in the present.

The example table provided here is written from the outpost's perspective, but a GM could just as easily use it to interpret recent past events for the native neighbors of the place. Some might pair in cause-and-effect matching, like a plague striking the natives that later allowed the outpost to expand.

Accident: Some unexpected failure of tech or judgment caused an accident that almost wiped out the outpost. Irreplaceable tech may have been lost, or the colony might have been left with a dangerous zone of rubble or radioactivity as a legacy. Local neighbors may have suffered as well.

Boom: Something around the outpost proved to be much more valuable than anyone had expected. Many colonists became rich, and many outsiders may have rushed to join the boom. It's probable the resource has petered out by now, though echoes of former money might remain.

Capture: Either the natives or some offworlder power seized control of the outpost for a time. They may have been driven off by a reprisal force, or they may have abandoned the site after extracting any value from it, or they may still have more or less control over it.

Expansion: The outpost swelled in size or population at one point. This may have been a planned increase in size funded by their patron, or some outside event may have driven many newcomers to them.

Hazard: Some unforeseen peril of this planet almost destroyed the outpost, and hasty measures had to be taken to save those who remained. The peril may be gone now, or it may be a persistent threat to the place.

Isolation: Most outposts have long spells between visitors, but this one had an abnormally long period of isolation. It may have been forced to hide its existence, or some quarantine might have been imposed on it. It's likely very self-sufficient, but may have peculiar cultural norms compared to the wider sector.

Pirates: While most outposts have at least some anti-space defenses, sufficiently cunning or powerful pirates can still mount raids on a colony. This outpost was scarred by a serious pirate attack, and they may have had to cut unsavory deals with their tormentors in order to ensure their own survival.

d12	An Event that Happened
1	Accident , one that almost wiped the outpost
2	Boom , great wealth from a local source
3	Capture , seizure by some outside power
4	Expansion , a sudden increase in size
5	Hazard , a natural hazard almost killed them
6	Isolation , an unusual period of solitude
7	Pirates , with a raid launched by offworlders
8	Plague , a sickness that caused dire harm
9	Privation , a lack of some critical resource
10	Raiding , natives attacking the outpost
11	Regression , loss of important usable tech
12	Strife , with violent internal dissension

Plague: While most alien worlds don't have a microbiome capable of infecting humanity, there's always room for ugly surprises in some steaming chromatic jungle. Some vicious alien plague ripped through the outpost, and it may have had dramatic effects unlike those of normal Terran sicknesses.

Privation: Planning was in error, or supplies were spoiled, or raiders may have plundered critical resources. Whatever the cause, the outpost suffered a period of dire want, with critical supplies lacking for most of the population. Grim customs may have taken root during this time in order to ensure that at least some of them could survive.

Raiding: Some native neighbor attacked the outpost or made a habit of ambushes. Bad blood and memories of lives lost doubtless cloud relationships between the outpost and their local neighbors. Some faction of the natives might still be determined to raid the outpost, out of greed, resentment, or the glory of defeating the alien invaders of their world.

Regression: Most outposts are equipped with the tech appropriate to their patron power, but this site somehow lost their ability to use important advanced technology. They may have had their equipment destroyed by a native fungus or environmental hazard, or it might be that using it is somehow too dangerous under local conditions. They've been forced to fall back on more primitive technology, and may not be much in advance of their low-tech native neighbors.

Strife: The outpost was once rent by violent civil conflict, with factions or parties willing to shed blood to advance their own goals. The losers of the strife may have been wiped out, or exiled elsewhere on the planet, or grudgingly re-admitted to the society, but it's inevitable that the conflict should have left its mark on the local society and government.

THE OUTPOST'S GOVERNMENT

An outpost rarely has a sophisticated governmental hierarchy, and few of them are given to democratic or conciliar rule. Small research posts or wildcat mines might have nothing more than a senior researcher or mine foreman to settle disputes and direct daily labor. Whether a colonial grandee or a grubby elder, however, the PCs are likely to need to deal with them sooner or later.

How is the outpost ruled? Who makes the decisions for the place, and what power centers do they need to keep happy about it? Is theirs a simple, patron-imposed autocracy with some senior director in charge, or is the outpost more of a frontier democracy? It's worth remembering that just because the patron *meant* for a place to have a particular kind of organization, there's no guarantee that the actual facts on the ground will agree with that plan. The CEO that they think is in charge might be held as a literal hostage by the mine foreman who actually has the allegiance of the laborers.

How were they chosen? This is the second question you ask, because most of their interactions with the PCs will be colored by this. How did the outpost's leader get their job? Were they appointed by the outpost's sponsor? Were they chosen by the residents? Did they usurp their position by virtue of force or cunning? While they may have perfectly fit the criteria of their appointment, those criteria may not be the best fit for the reality of their duties. Some outposts may have several leaders or a town-hall meeting style of direct democracy, but most patrons will want a single viceroy to direct.

How do they maintain their rule? Why does anyone pay attention to them? What kind of muscle, legitimacy, or suasion do they have to make the other citizens pay attention to their wishes? With the homeworld a long drill distant and the possibility of months or years of solitude between serious inspections, an outpost ruler needs to be able to control the place with their own resources or capabilities rather than re-

lying on borrowed might from their sponsor. They also need to possess enough basic competency or technical expertise to actually run the place, as there's rarely any spare margin for arrant stupidity in an administrator. Of course, the tale of an incompetent director bringing an outpost to ruin can make for the start of a number of adventures.

What do they want from the PCs? Most outposts are isolated enough that a band of freebooters with access to a starship is a resource worth exploiting. An outpost leader almost certainly has a wide selection of problems that could be usefully addressed by inessential strangers, particularly strangers who may need some favor or permission from the leader. Treat their problem as an excuse to drop a nice, tasty adventure hook in front of the PCs.

What's their name and appearance? Once you know how they got their job, how they keep it, and what they might want from the PCs, you can top them off with a name and suitably memorable appearance or manner. It can be useful to gesture at problems or capabilities with this appearance. For example, if the administrator keeps control by a group of heavily-armed security personnel, hint at it by giving him a couple of those goons as constant companions.

It's not uncommon for directors or other leaders to have a different uniform or some token of office carried visibly when meeting outsiders. Within the outpost, the population is usually small enough that no such reminders are necessary. Sometimes these tokens of office can have a practical utility, especially if they're some locally-discovered pretech relic confiscated by an earlier leader. Occasionally their real purposes may be unsuspected by their current owners.

The one-roll tables on the following pages can be used to give some fast results for you to polish into something acceptable, or to be used as inspirational prompts for your own creations.



FORMS OF GOVERNMENT

Given the usual size of an outpost, it's unlikely to have a particularly sophisticated form of government. A few thousand workers in the same general place and with the same shared goal can usually expect to be overseen by a single administrator and a modest apparatus. In many cases, an outpost's patron has no desire for their employees or colonists to have any more self-rule than their circumstances require.

Of course, with uncounted light-years between the head office and the reality of a steaming alien jungle, there's a lot of room for the corporate handbook to get modified. Outposts will tend to respond to the pressures of their environment, and an administrative form that doesn't suit their needs will be rapidly converted to one that does. In other cases, powerful colonists or ambitious natives can force the issue, compelling a de-facto change of government even if the quarterly reports are still sent under the old letterhead.

Anarchy: The outpost has no recognized leaders, with the citizenry free to do as they see fit. Something has prevented the usual progress to the tyranny of brute force; the locals might all be too reliant on each other to risk violence, or the outpost might be small and new enough not to have developed that far, or an immediate external threat might be forcing a postponement of internal strife.

Bureaucracy: The outpost's governance is divided up among various bureaucratic departments related to its original function and patron. The heads of these departments have control over their own spheres of influence, and so control hinges on convincing or forcing them to cooperate towards some purpose.

Feudal: The outpost and its valuable or resource-generating elements have been shared out among various leaders, with one of them acting as a first among equals. The central authority's power is limited, and they usually need to convince at least some of their subordinates to willingly cooperate in their plans.

Hereditary: The leadership rests with the heirs of a founding family or original leader's clan. In some cases this is because of a belief in their natural suitability for the role; in others, it's because the other major powers in the outpost don't trust its rule to any of their rivals.

Kleptocracy: Pirates or criminals have taken over the outpost, and now it serves chiefly to enrich them or provide a refuge against the law of other systems. Whatever nominal form of government is present is actually wholly subordinate to the rule of crime bosses or pirate captains.

Mayoralty: The outpost as a whole votes or otherwise chooses its leadership. Small outposts might have only a single elected mayor, while larger ones might likewise vote on other important offices.

di12	Types of Outpost Government
1	Anarchistic cooperation or lack thereof
2	Bureaucratic cooperation by mandarins
3	Feudal cooperation between gentry
4	Hereditary rule by a founding family
5	Kleptocracy of a criminal or pirate elite
6	Mayoralty based on a vote of the citizenry
7	Native rule imposed by force or pragmatism
8	Technocracy based on a vital skill
9	Theocracy based on religious principles
10	Tribal affiliation with ideologies or clans
11	Tyranny of brute force by a warlord
12	Viceregency imposed by the patron

Native Rule: The natives of this planet actually control the outpost. They may have such an overwhelming strength that the nominal leader is totally subject to their wishes, or the outpost's mission might be so dependent on native cooperation that it's the locals who actually run the show.

Technocracy: The outpost's mission relies on a particular skill, and those who possess the skill are the ones who make the decisions for the settlement. An asteroid mine might defer all decisions to the mining engineers, while an artist's colony might leave administrative choices to the most celebrated artists. Skill in this field may not correspond well to actual administrative competence.

Tribal: The outpost is made up of several extended families or clans, and decisions are made by the patriarchs and matriarchs of these groups. Not all tribes are related strictly to blood association, however, and ideological, religious or planetary-origin distinctions might also birth tribal behavior.

Tyranny: The outpost is run by the person with the most force at their disposal. They are not necessarily any crueler, more unjust, or more rapacious than any other administrator, but they obtained their power by virtue of violence, and are usually perfectly willing to keep it in the same way.

Viceregency: The patron group makes all serious strategic decisions for the outpost, but they've dispatched a viceroy to handle implementation and day-to-day decision making. If this viceroy dies or is incapacitated the outpost will have to wait for a replacement to be sent before they have a legitimate administrator once more.

WHO IS IN CHARGE?

Most outposts are small enough that the PCs can expect to be able to deal with the actual leader of the place. Only in the largest or most security-minded outposts will they have the run the gauntlet of lesser functionaries and minor clerks that usual interpose between freebooters and the executive officers of a world.

With that in mind, it's necessary to establish the important facts about the leader, because the party will almost certainly be talking to them sooner or later. The tables on this page offer suggestions for hammering down the most important elements of their rule, and give some ideas for memorable quirks or personal problems that the PCs could leverage for their own ends. Remember that the purpose of this leader is to provide adventure hooks, job rewards, or antagonistic challenges to the PCs, and write them accordingly.

d6	The Popular Attitude About Them
1	No strong feeling; an adequate leader
2	A few hate them, but most don't care much
3	A recent act caused strongly divided camps
4	They have a lot of trust or fearful respect
5	They're disliked or disrespected by many
6	They're deeply admired or greatly feared

d8	How Were They Chosen?
1	Popular sentiment that they were the best
2	Secret betrayal of the former leader
3	Imposed by brute force by a faction
4	Careful political dealmaking raised them
5	They bought it with their own wealth
6	Kingmakers thought them a pliant puppet
7	No other viable candidate wanted the job
8	Random chance pushed them into it

d10	What's Their Strongest Tool of Rule?
1	Extremely good at efficient administration
2	A cultish popularity in some circles
3	Legitimated by the local religious leaders
4	They have powerful offworld backers
5	They're personally terrifying or violent
6	They have strong ties with native rulers
7	Tight control of a critical outpost resource
8	They're the only one with some critical skill
9	They've set their rivals at each others' throats
10	They have a powerful supply of muscle

d4	How Long Have They Ruled?
1	They've just been put in power
2	A while, but still uncertain of some things
3	They've been in charge for a long time
4	So long that they're getting complacent

d12	What Do They Want From the PCs?
1	Deal with a troublesome rival
2	Resolve an issue with the natives
3	Acquire some illicit but critical contraband
4	Get some vital spare parts
5	Pay them a large sum of credits
6	Smuggle local goods to a particular world
7	Recover something the natives got or stole
8	Rescue a citizen who's gone astray
9	Clear out a natural hazard or hostile tribe
10	Sabotage some other offworlder effort
11	Take somebody far away, willing or not
12	Stop a current native aggression

d20	Quirks of the Leader
1	They hate everything about this place
2	A trait about them deeply offends the natives
3	They have a secret, deeply impractical goal
4	They're desperate for offworlder socializing
5	They're addicted to a local substance
6	They've fallen in love with a native
7	They want to quit and hand over leadership
8	They're smuggling goods offworld
9	They're a puppet of a secret leader here
10	They excessively idealize the local natives
11	They're besotted with native pleasures
12	They're exhausted from excessive demands
13	They're a native who made good somehow
14	They tell people what they want to hear
15	A rival has substantial blackmail on them
16	They're zealous about a faith or ideology
17	They hate it here and take it out on everyone
18	They have grandiose dreams and plans
19	This is a punishment post for a prior failing
20	They complain endlessly about their job

WHO IS OPPOSING THEM?

Not every outpost has an active opposition to the leadership. In some of them, the chief might be so well-liked or apathetically accepted that there's no real hostility to their rule, nor any significant challengers to their position.

Other times, however, you need a counterweight to their authority. If the leader is exceptionally sympathetic and friendly toward the PCs, you might need an antagonist to provide the problems that the PCs are needed to resolve. Conversely, a cruel tyrant of a ruler is likely to send the PCs looking for some rebel leader to help them achieve whatever ends the outpost ruler is preventing them from accomplishing.

It's not necessary to have just one opponent. It may be that there are several would-be usurpers to the outpost's rule, all of whom may be mutually hostile.

d6	Who Are Their Main Supporters?
1	Natives neighbors or offworlders
2	Common laborers and the lower classes
3	Religious believers or ideologues
4	The gentry or foreman classes
5	A particular clan or tribal group
6	Criminals, ruffians, and social outcasts

d8	Why Are They Supported?
1	They're very free with financial help
2	They have tremendous personal charisma
3	They've the right marriage or kin relations
4	They have a very attractive ideology
5	They're thought just and noble-minded
6	People are personally afraid of them
7	They're viewed as the defender of a group
8	They're thought far more capable

d10	Why Haven't They Taken Over?
1	They think rule will drop into their hands
2	They tried, but were foiled recently
3	They're setting the ruler up to fail badly
4	They're uncertain of their own strength
5	They still need to obtain a critical support
6	Rivalry is more profitable than success
7	They don't trust their own subordinates
8	Their supporters have different ideas
9	The ruler is bribing them to inactivity
10	They're waiting for the right catpaws

d4	How Known Is Their Opposition?
1	They're thought a loyal supporter.
2	They seem to be uninvolved in affairs
3	Their enmity is known in a disguised ID
4	Everyone knows they're a bitter enemy.

d12	What's Their Latest Action?
1	Murder of a major supporter of the ruler
2	Theft of some valuable tech or resource
3	Subversion of a minion of the ruler
4	Sabotage of a plan dear to the ruler
5	Jailbreaking of a rebel supporter
6	Propaganda against the leadership
7	Support of a downtrodden group
8	Negotiations with potential native help
9	Contact with offworlder allies
10	Obtaining use of a ship or other costly tech
11	Acquisition of a dangerously skilled ally
12	Construction of a profitable enterprise

d20	Quirks of the Rival
1	They're deadly sincere in their ideology
2	They're the heir to a former rebel leader
3	They feel morally licensed to do anything
4	They crave offworld support
5	Outsiders are expendable pawns to them
6	They irrationally despise some local group
7	They're actually an agent of the ruler
8	The rivalry is purely a grift to them
9	They're utterly narcissistic and self-centered
10	Their ideals are hopelessly impractical
11	They actually should have been the leader
12	They're a rival out of a sense of grim duty
13	The rivalry is an excuse for a personal grudge
14	They feel desperately overwhelmed
15	They have a bad case of paranoia
16	Their cause is precisely half-justified
17	They don't want to harm the ruler
18	They have access to psychic resources
19	They need to be ruler or else face a dire fate
20	They really are the de facto ruler here

THE OUTPOST'S SITE

You know what the outpost means and you know who's in charge. But what about the actual physical site of the outpost? What does it look like, how do the PCs interact with it, and what's going on there that would be interesting, threatening, or profitable for them?

It's not necessary to make up a map for every borderworld outpost your PCs land at. The specific details of streets or buildings can be extemporized if they turn out to be important for a firefight or a heist. It is, however, necessary to establish a few detailed points about the place.

What's the overall look of the place? Write four or five sentences describing what the place looks, sounds, and smells like. What are the buildings constructed of, and how big are they? Does the place have walls and other defensive fortifications, or is it unnecessary here? Is it in good repair and do the locals have the time, energy, and civic-mindedness to keep it up?

How elaborate is the starport? There's some place for incoming traffic to land, whether that's no more than a flat patch of bedrock outside the outpost or a titanic capital-class docking port at some deep-space colony hab. The more traffic the outpost gets, the more elaborate their starport is going to be, and the more security-conscious or restricted the place is, the more tightly they're going to contain visitors until they've been duly inspected and approved. Almost any starport worthy of the name should have fuel for sale, but not all of them may have much in the way of repair facilities or the ability to patch up seriously damaged ships. Those that do may restrict their availability based on the party's ability to do some favor for the leadership.

What adventurer-relevant goods and services are available here? An outpost may be making tremendous scientific strides with their research into crystalline mineral formation, but the average adventuring party is more interested in whether or not they can buy guns there. What kind of lodging

facilities are available for outsiders? Do they have good medical care here? Does their work here involve particular tools or equipment that the PCs might want to buy, including tech that might be illegal on more developed worlds? Can the PCs get interpreters and guides for handling the planet's natives or natural hazards?

What will they do if the PCs make trouble? Adventurers are notorious for their quick resort to violence, theft, and general chicanery, so what will the locals do if the PCs suddenly start pulling guns on the citizenry? What resources do they have for maintaining local order and responding to sudden crises? The outpost almost certainly has some sort of ground-based anti-ship defenses if they're not to be vulnerable to extortion from every far trader with a popgun on their ship, and all but the most impoverished colonies will have nuke snuffer fields active and quantum ECM in place around critical sites. It may be that a heavy frigate full of power-armored PCs could smash whatever defenses the colony possesses, but the place ought to be able to stand up to the ordinary threats it can expect to encounter regularly.

What do ordinary locals look and act like? For your future convenience, make up three ordinary NPCs with the tools in this section. Each of the three should have a connection to the unique or specific characteristics of the outpost, and one of them should be the first NPC the PCs encounter. If the outpost is a crystal mine, for example, you might make one of the NPCs a crystal miner. If they're having problems with the natives, then maybe the NPC spaceport manager recently lost a loved one to a native raid and is thirsting for revenge. The point is to make these NPCs engaged with this specific outpost's situation. When you create adventures in the outpost or need stock NPCs to fill in the blank of some social interaction with the PCs, you can grab one of these three to stock your situation.



ARCHITECTURAL AND LAYOUT STYLES

It's not generally necessary to map an outpost or set down hard details of its layout, but it's often useful to have a few general descriptors of the place and a vague idea of how the place looks. These tables can be used to generate a few characteristic elements for when you need to describe some dive bar or a bustling colonial street scene.

These results will need to be tweaked based on the environment of the outpost, as a settlement on an airless asteroid won't have the same concerns with weather and native relations that a colony on a former garden world might have. In the same fashion, an outpost on a planet with no breathable atmosphere or a temperature high enough to melt lead is going to be much more tightly buttoned than one that was established on a more clement planet.

d6 Popular Building Shapes and Outlines

- 1 Blocky cubes or rectangular solids
- 2 Domes and geodesic structures
- 3 Wings encircling a central courtyard
- 4 Cylindrical towers and spires
- 5 Organic, soft-edged masses
- 6 Skeletal struts with enclosed areas

d8 Color Palettes

- 1 Bare concrete and metal
- 2 Solid masses of bright colors
- 3 "Natural" wood and earth textures
- 4 Neutral browns, grays, and whites
- 5 Stylized cultural patterns and glyphs
- 6 Patron-related colors or symbolism
- 7 Camo hues to blend into the surrounds
- 8 Sci-fi white and crystal motif

d10 General Outpost Layout

- 1 Rigidly-enforced grid of streets
- 2 Clusters of task-related buildings
- 3 Random organic growth from a core
- 4 Ultra-compact, easily-defended zone
- 5 Residential, commercial, industrial zones
- 6 Organized core with a shantytown suburb
- 7 Long and narrow against a local feature
- 8 Status or role-based sub-districts
- 9 Roll 1d8 twice; both styles exist together
- 10 Roll 1d8 twice; it was rebuilt once before

d4 How Vertical is the Architecture?

- 1 Single-floor or partly subterranean
- 2 A few multi-story buildings
- 3 Densely built-up multifloor structures
- 4 A single major structure or arcology

d12 What's A Local Infrastructure Problem?

- 1 The sewage system keeps going haywire
- 2 Power generation isn't as reliable as wished
- 3 It's hard to maintain a comfortable climate
- 4 A type of native weather overwhelms it
- 5 The techs aren't competent to handle it
- 6 Not enough laborers for the dirty work
- 7 Water supplies are impure or scarce
- 8 Roads or passages are often blocked
- 9 The natives keep salvaging valuable parts
- 10 The outpost's too big for its infrastructure
- 11 They're cannibalizing gear to keep it up
- 12 A local danger tends to break in at times

d20 Quirk of the Place

- 1 Building connections are all tunnels
- 2 The environment is decaying things quickly
- 3 A local fungus or plant invades everywhere
- 4 It's always much too humid, hot, or cold
- 5 Propaganda or ideological posters all over
- 6 There's a constant noise in the outpost
- 7 Passcards are needed to pass every door
- 8 Ugly native vermin creep in everywhere
- 9 The local environment has a certain stench
- 10 Regular minor earth tremors
- 11 The wind is strong and incessant
- 12 A lot of local tech is jury-rigged into working
- 13 An abnormal number of security cameras
- 14 Only a very few places for socializing
- 15 Very visible monument to their dead
- 16 A minor local sickness is ineradicable
- 17 The atmosphere filters don't work perfectly
- 18 A ruined facility that went very wrong
- 19 Branch of some interstellar organization
- 20 A section where native locals congregate

AN AVERAGE OUTPOST CITIZEN

Most outpost citizens will be drawn from the same founding stock. On extremely ethnically diverse worlds, this might result in a grab-bag of different looks, but most planets have homogenized to a significant extent over the past few centuries, and it's likely that most of the locals share the same basic physical appearance. The tables below can be used to give you a baseline.

Of course, isolated frontier outposts are a notorious refuge for castoffs, adventurers, and other interstellar ne'er-do-wells, so it's not impossible that some of the citizens look very different from their peers.

Note that isolated outposts are also prone to developing their own local customs and habits. Some of the traits or styles generated may be unique to the outpost, ones established by past events, recent trendsetters, or some outside influence.

d6	Most Common Hair Style
1	Bald or shaved in specific patterns
2	Close-cropped or aggressively short
3	Masses of tight curls
4	Relatively short, straight hair
5	Long wavy or curly locks
6	Long straight hair, whether done up or loose

d8	Common Skin Color Ranges
1	Extremely dark
2	Mahogany, dark brown, coffee shades
3	Bronze, ruddy brown, dark golds
4	Olive, light browns, tanned shades
5	Golden, sallow, or old ivory
6	Pinks and pale ivories
7	Near-albino pallor
8	A color tint unknown to modern humanity

d10	Common Natural Hair Colors
1	Snow whites or platinum blond shades
2	Yellow gold, straw yellow, wheats
3	Pale or medium browns
4	Dark browns or mahoganies
5	Bright reds, crimsons, and scarlets
6	Dark reds and ruddy browns
7	Very dark near-blacks and browns
8	Ravenfeather black, blue-black
9	Roll 1d8 twice; a natural pattern of both
10	Abnormal but gengineered-natural color

d4	How Big Are Most Of Them?
1	Short, whether broad-shouldered or lithe
2	Same build as most humans in this sector
3	Average height, but stocky or slender
4	Unusually tall, whether thin or bulky

d12	Common Outpost Casual Wear
1	Loose-fitting robes
2	Long-sleeved tunics and trousers
3	Overalls and repurposed work clothing
4	Tunics and skirts or kilt-like garments
5	Close-fitting robes and wraps
6	All-encompassing shrouds and veils
7	As little as the environment allows
8	Snug-fitting jackets, shirts, and trousers
9	Armor-like protective gear
10	Baggy, loose-fitting jumpsuits
11	Uniforms related to social or religious ties
12	Skin-tight jumpsuits with work belts

d20	Common Local Style, Habit, or Quirk
1	They're very fond of perfumes and scents
2	Hair styles are exceptionally impractical
3	A particular common color is offensive
4	Everyone carries visible religious tokens
5	Their voices are unusual in some way
6	Body dyes and patterns are popular
7	Minor or major ritual scarring
8	Social sex segregation is common
9	They hate to say "no" directly to people
10	They like lights everywhere, even on clothes
11	Everyone has large tattoos of some sort
12	They're unusually bad with other languages
13	They love to gamble with each other
14	Some aggressive body mod is common
15	A certain skill is common and esteemed
16	Older figures are treated with great respect
17	They use a lot of recreational drugs
18	Being armed is a cultural requirement
19	They're all crazy about a particular hobby
20	Pet native animals are common to many

SERVICES AND COSTS

Aside from being a source of adventure hooks, the other primary use of an outpost to most PCs is as a source of resupply when far from an inhabited world. When establishing the details of the outpost site itself, you should go down the list below and note the presence or absence of the given service, including any special local costs or favors needed to get access to it.

Common Types of Outpost Services

Carousing: They have fun ways to spend money

Crime: There's an accessible criminal network there that can provide illegal services and favors

Gear: They offer non-lethal TL4 equipment

Information: They can provide valuable info about the world or some matter the PCs care about

Medical: They can do more than emergency care

Recruits: Some of the locals can be hired on

Refueling: They can provide starship fuel

Ship Repair: Fixing damage and doing maintenance

Transport: They can find local transport here, or hire a passing ship captain to take them offworld

Weapons: They'll sell TL4 weaponry to outsiders

An outpost might have the physical ability to grant a particular service, but whether it does or not may be up to the PCs' inducements. Most outposts will be willing to sell refined ship fuel, if only to ensure that the interlopers can leave, but the more reclusive ones may need some persuading before they're willing to part with local specialty products or valuable supplies.

If you don't want to make a full-fledged adventure out of persuading the locals to cooperate, you can use the table below to come up with a price for its access.

d6	What Do They Want?
1	Money. There's an "administrative charge" that the service provider is likely to set to whatever sum the PCs are able to pay.
2	Crime. Whoever's in charge of access to the service wants a minor crime done that outsiders have a better chance of pulling off.
3	Time. They'll make it available to the PCs, but it's going to take a week or two. Might as well find something to do around here...
4	Help. The service chief has a problem, and if it was solved things would go faster.
5	Somebody. The person who can give access is missing and needs to be found.
6	Grudge. Using this service somehow makes an enemy of another service's chief.



At times, an outpost that seems like it should offer a particular service can't or won't make it available to the PCs. Usually the reason for this is a problem that the party can resolve as part of a larger adventure or as a short interlude during an otherwise-occupied play session.

d8 Why Can't They Offer the Service?

- 1 The building is infested by alien vermin, contaminated by a recent accident, or otherwise rendered inaccessible at the moment.
- 2 The patron doesn't want them to, and there's an inspector or other spy around at the moment who might detect such dealings.
- 3 The person who can provide it has some kind of personal grudge against the PCs, either for something they did or something they are.
- 4 The kind of tech or support they offer is somehow incompatible with what the PCs have or need, being designed for a different recipient.
- 5 They have a sudden and unanticipated local need for large amounts of the good or service.
- 6 They offer it, but it's of such a low standard or quality that it risks doing more harm than good.
- 7 The facility that provides it has been taken over for some other, more immediately-pressing need of the outpost.
- 8 They could, but they're just too busy or too uninterested in helping to go to the trouble of offering it to the PCs.

TOOLS OF THE PIONEERS

This section of *Distant Lights* includes prices and guidelines for building an outpost, whether a crew-sized pressure dome to pack in the back of the ATV or a full-scale offworld colony capable of housing hundreds of people for an indefinite amount of time. Players can use these rules to plan out their own interstellar waystations, while GMs can use the prices and scales as a reference for colony-based adventures.

Very large colonies with active industrial bases and a population big enough to form an independent economy aren't covered by these rules; they're meant to handle the small offworld outposts most likely to be relevant to an adventure. GMs seeking instruction on how to handle those grander undertakings and bolder trading efforts can find rules for it in the *Suns of Gold* supplement.

Reasons for an Outpost

It's self-evident why a megacorp might want to plant an outpost on some outcropping of rare minerals or on an Eden-world with exotic genetic resources, and there's also little to wonder about some minority group who might choose to voyage to an untouched world of their own. For PCs, though, it can be harder to explain why they would choose to establish a base camp or outpost on some distant planet. For a PC party, the reasons usually boil down to a few common motivations.

It's a forward base. They intend to do a lot of things on this planet or become deeply involved with the local power structures, and it's not practical for them to just live in their ship. They may have too many supporters or hirelings to comfortably keep them all ship-side, or they might not even have a ship of their own and must rely on chartered craft. Building a forward base lets them securely amass more cargo, people, and facilities than they can fit on their free merchant.

It's a refuge. They've got enough enemies that they need to have a well-fortified home somewhere too far away to be

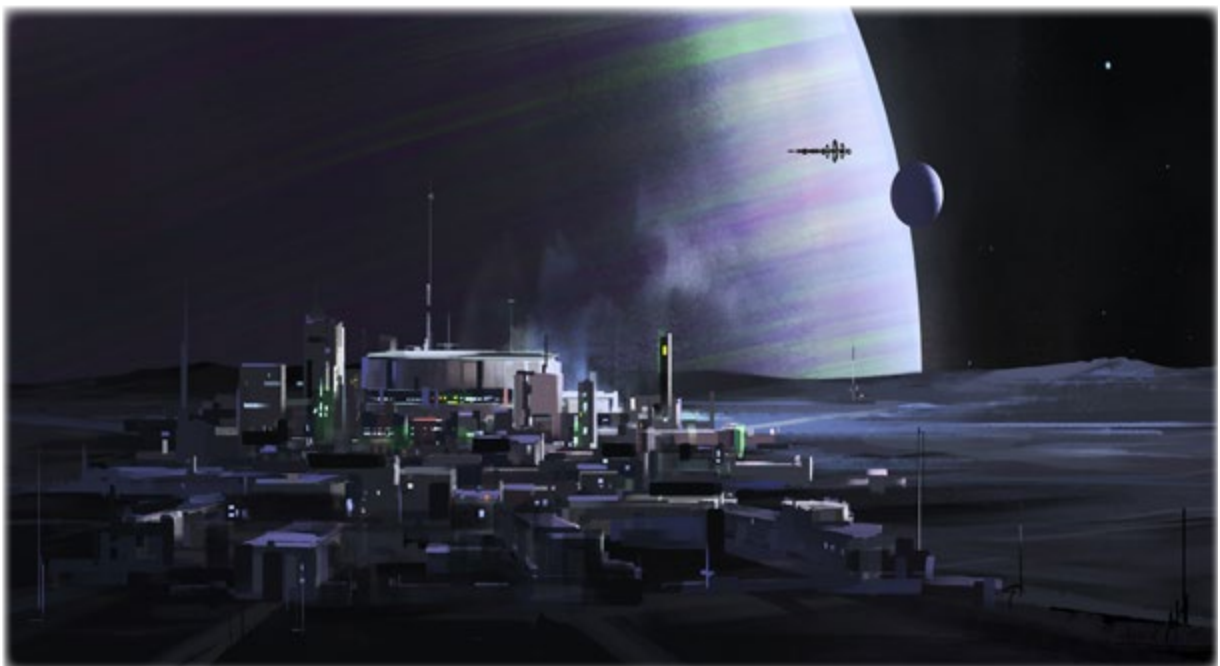
easily reached by assassins or hostile governments. An outpost planted on some asteroid in the middle of cosmic nowhere has much to recommend it for security, while a base camp erected in some steaming alien jungle isn't likely to be found by anyone who doesn't know where to look. In the same vein, this refuge might be needed for allies or friends who wouldn't be safe elsewhere in the sector.

It's a profit center. There's something there that's valuable and worth extracting or processing, but the PCs don't want to have to spend months doing their own work there. They need to hire a work crew, establish an outpost, and then profit by their capital investments. On a suitably uninhabited planet, this might make the PCs the de facto rulers of the world, until some sector hegemon's cruiser shows up with a marine detachment ready to demonstrate their heavily-armed friendship.

It's a prison. PCs sometimes deal with NPCs that they can't or won't kill, but can't afford to have running loose. Dropping these captives off at a pressurized building surrounded by radioactive acid clouds is a very good way to ensure they don't wander off without somehow getting outside help.

It's a palace. Some players just really enjoy having a place to call their own and fitting it out with all the luxuries and conveniences that a successful interstellar freebooter might desire. Such an outpost might turn out to have some very practical utility when the group needs to get away from things for a while, but it's the pleasure of having the place that matters most to its builders.

As the GM, pitching the benefits of a privately-held outpost is a good way to help bleed off credits from a party that's just made an exceptionally good score. Every additional bit of property or connection they establish with your setting is just one more hook to help keep them engaged and interested in the campaign.



Outpost Statistics

Truly zealous planners might enjoy tracking specific tons of food, liters of water, tallies of atmospheric filter masks, and other critical outpost supplies. Most tables aren't really interested in such a finely-grained simulation, however, so the guidelines in this book focus on only four scores that really matter to a colony.

Population is the total number of people that the outpost can comfortably house. This number might be considerably increased for a short period by hot-bunking or shared housing, but for long-term stability, this number reflects a safe total. This number also reflects the infrastructure necessary to maintain a pressurized environment or safe atmosphere on worlds where the air isn't breathable, or on asteroid bases that lack any atmosphere at all. This infrastructure is extended to any other facilities the colony might erect, so pressurizing the factories and garage bays of the outpost isn't something that has to be tracked separately.

Staff is the number of people necessary to maintain the outpost's normal functioning. A colony might be able to keep going on as few as half this number in the short term, but exhaustion, neglect, and lack of skills can often end up doing substantial damage. Each outpost facility has its own minimum required staff, and during a personnel shortage it might be necessary to shut down particular facilities so that power generation and hydroponics can stay fully-functional. Staff generally need to be paid, and while they'll work to keep themselves alive they will not be very happy with their patron unless recompensed for their labor.

Power is the total power required to keep the outpost facilities operational. Generators add it, and most facilities cost it. An outpost that's running short on power will need to shut down some facilities in order to maintain critical functionality. A hab block or other pressurized residential area that runs out of power won't be able to maintain a safe atmosphere for long.

Workspace is functional space dedicated to industrial or technical labor. A hab block may be beautifully optimized for sheltering people, but it hasn't got the open work bays, heavy-duty wiring, reinforced construction, or sterilization seals necessary for industrial or medical pursuits. Some facilities provide workspace as part of their construction, while many non-residential facilities cost a certain sum of it.

Building an Outpost

To build an outpost, the necessary facilities have to be brought to the site and erected. Each facility lists its required cargo tonnage, and a ship or other conveyance will need to have that much free space in its hold to pack the facility. Once it gets to its destination, laborers will need to unpack it and use its assorted self-assembly functions and construction helpers to get it in operating order.

A single laborer can build 2,000 credits worth a facility in one day. Thus, a block of Basic Residences that costs 25,000 credits would take thirteen days to assemble for a lone laborer. Building a fifty-million-credit military starport would require a considerably larger workforce to get the job done. On most outpost worlds, it's an accepted social convention that citizens who don't already have other jobs to attend will help erect new

outpost buildings. In less altruistic settings the locals might need to be paid to help out, or labor bots might need to be imported to help do the work.

When constructing facilities on hostile worlds or in the airless void of space, a starship or adequate temporary housing will need to be available to keep the laborers alive while they put together their new home. Most such workers can tolerate rougher temporary housing if better is in the offing.

Acquiring staffers for an outpost will depend on the PCs' own ingenuity and contacts. Good pay, safe work, and the assurance of an eventual trip home is usually enough to draw capable hires, but more difficult circumstances may require bigger inducements. It's not unknown for outright slave labor to be used in some remote outposts.

Maintaining Outpost Facilities

Advanced TL4 buildings designed for colony usage are meant to be tough, simple, low-maintenance structures. Even so, the very simplicity that makes them so durable means that regular maintenance and upkeep is often needed to keep things in working order.

An outpost requires 5% of its total facility cost each year in spare parts in order to keep it running. Hyper-advanced facilities like the Gravity Generator that have no listed cost are too advanced to require any maintenance... or at least, any maintenance that a TL4 society could provide. Some worlds may be able to produce their own spare parts in a local Industrial Processor, while others may be forced to rely on offworld shipments of vital spares.

An outpost that doesn't get this maintenance runs an increasing chance of having significant facility failures. Each month the backlog goes unpaid, roll 1d6; on a 1, a random facility has gone offline and adds 10% of its cost to the backlog, as deferred maintenance has resulted in serious damage.

Actually *performing* the maintenance work is assumed to be part of the staff duties of each facility, so no extra fees need to be paid for the labor.

Outpost Payrolls and Supply

An outpost usually has workers who must be paid and mouths that must be fed. If the outpost is capable of producing its own food supply these costs can be cut substantially, and an Industrial Processor dedicated to the production of basic necessities can likewise lower the supply costs. In very few cases will the outpost denizens agree to work without pay, however.

Each person in the outpost requires 5 credits worth of food each day amounting to one kilogram of edible mass. A Micro-Greenhouse, Robotic Farm, or other food-producing facility can eliminate this cost. Otherwise, the outpost's patron will have to buy these supplies and ship them in as cargo. Secure storage is advisable, as such stockpiles are ideal targets for hungry locals and avaricious native life forms.

A person who goes without food will be too weak to work in three days and dead within two to four weeks, on average. On particularly harsh worlds where no supplementary scavenging or hunting is possible, death may come sooner. On worlds where hunting and gathering has a hope of preserving life, a person might be able to hold out indefinitely, though



they'll be far too busy surviving to do any other constructive work.

Each person in the outpost requires 5 credits worth of miscellaneous supplies each day, including clothing, consumer goods, entertainment, medicines, and other goods necessary to keep them comfortable and happy. An Industrial Processor dedicated to building such common goods can supply the locals, if the native resources are suitable for processing.

A person who goes without supplies will be increasingly upset. While it's unlikely to kill them in most environments, prolonged lack of necessary goods can provoke rebellions or abandonment of the colony, and can result in facilities shutting down due to strikes, sabotage, or slowdowns.

An outpost staffer must be paid an average of 30 credits a day, exclusive of food and supply costs. This is an excellent wage on most TL4 worlds, but the remote location and regular dangers of outpost life require a comparatively high pay for even very simple jobs. Expert technicians and medical staff might be paid more than this, and janitors receive less, but an average cost of 30 per staffer per day is a working assumption. Some staffers might work for considerably less, or even nothing at all if they have strong ideological motivations or are "encouraged" by armed overseers.

Staffers who are not paid will not work, unless their work is necessary to keep themselves alive. In that case, the outpost's patrons can expect to receive a very hot reception when they next visit. Slave labor runs the risk of rebellions and sabotage, though a population trapped on a hostile world with no prospect of escape can easily become demoralized.

Local residents who are not staffers need not be paid, though they'll still need food and supplies to remain comfortable. They may have local jobs or sidelines supporting the staff members of the outpost, taking their earnings from their own work. Most colonies avoid housing too many of these surplus citizens when possible, but longer-term outposts must make allowances for family and those too old or young to labor.

Outpost Profits

A colony placed near a valuable ore lode or in some region of useful alien flora can turn its owners a profit, and indeed, many outposts are established for no better reason. Groups that want to trade with a native population or establish major industrial hubs can use the rules in the *Suns of Gold* supplement, while the rules below can be used for smaller-scale claims.

Profit begins with an Extraction Industry facility staffed by workers who pry the precious ores, rare plant extracts, or alien monster organs from the reluctant world around them. The exact nature of these goods is up to the GM, depending on what the world has to offer, but they can generally be divided into four different tiers of value.

Common resources are ordinary metals, chemicals, plant feedstock, and other materials that can be found almost anywhere. While they are flexible and versatile extracts that can often be manufactured into useful colony supplies, they aren't of much worth in the larger cosmos. A large colony might have some workers laboring to extract these materials simply to

ensure a local supply of clothing, tools, and other common essentials, even if it's more work and up-front expense than simply shipping them in from offworld.

Somewhat Valuable resources are uncommon ores, plant extracts with some special quality, or other substances that are seldom found elsewhere and have a practical use. Valuable resources usually have not only a practical use, but a very good one; the crystals here might make superb laser rifle emitters, for example, or the native leathers are amazingly resistant to heat. Precious resources are perhaps unique to this world, and are a critical component of some very useful or important good.

Each worker in an Extraction Industry digs out or refines so many credits worth of these resources every day. A day's work produces 10 credits of Common resources, 50 of Somewhat Valuable resources, 100 of Valuables, and 200 of Precious substances. Thirty days worth of output counts as one ton of ship cargo, so a ton of Common resources is worth 300 credits.

Many colonies simply pack up these goods and ship them out to be processed elsewhere. Larger or more sophisticated outposts can do post-processing work on these goods to make them more valuable, whether that consists of turning raw petroleum into finished plastic goods or refining exotic crystals into polished component lenses.

Industrial Processors can use these raw materials to create finished goods. Most processors are flexible enough to create a wide variety of TL4 goods, provided they have a plausible feedstock for them. A selection of Common ores and petrochemicals might justify making vehicles, colony supplies, common weaponry, or other useful but low-margin goods. A Precious strike of sun-vapor might only be good for fabricating the extravagantly expensive and addictive divinity drug the sector hegemon enjoys so much. Swapping production in a processor usually takes a week or so.

A single Industrial Processor staffer doubles the value of any material inputs they finish. Thus, a ton of common raw materials worth 300 credits could become a ton of energy cell components worth 600 credits. A single worker can produce up to 500 credits worth of finished products each day. Like extraction workers, a week's work-shift output averages to seven times this amount.

Once the goods are extracted or finished, they'll need to be stored until a trader lands to purchase them. Most facilities have enough room to store any reasonable amount of stock under adequate guard, though exceptionally precious goods might demand extra care.

The frequency of buyers will depend on the circumstances. If the PCs have made arrangements with a merchant cartel to make regular visits to their outpost, it might be a simple and predictable arrangement, albeit probably at a discount sale price. If they rely on free merchants and passersby, they might go far longer between sales. Some PCs might haul the goods themselves to a suitable market. If they want to get better prices than the base value of the goods, the GM can use the trading rules in *Suns of Gold* to cook up mercantile adventures and determine what their haggling can earn them.

Outpost Medical Needs

On a safe world, doing safe work, an outpost can usually get by on medkits and bed rest. There will be the occasional fatal sickness or tragic accident, but the power of advanced TL4 medicine can handle most day-to-day medical needs. When the planet is more dangerous, however, or staffers are undertaking risky extraction work or industrial labor, the consequences of a medical lack can be worse.

In such conditions, the GM rolls a 1d10 each month. A smaller die can be used for safer worlds, or a larger die or one with bonuses for more hazardous places still. That percentage of the staff will need serious medical attention, rounded up. Thus, if there were 50 workers and the GM rolled a 5, then 5%, or 3 people, would need care at a dedicated medical clinic or hospital. If the outpost doesn't have enough beds to support them, those not tended have a 50% chance of dying or being permanently crippled and unable to work.

Outpost residents are generally made up of people with a considerable appetite for risk, but consistent fatalities due to a lack of adequate medical care is likely to leave them restive. Such deaths might result in strikes or production slowdowns until more permanent medical facilities are built.

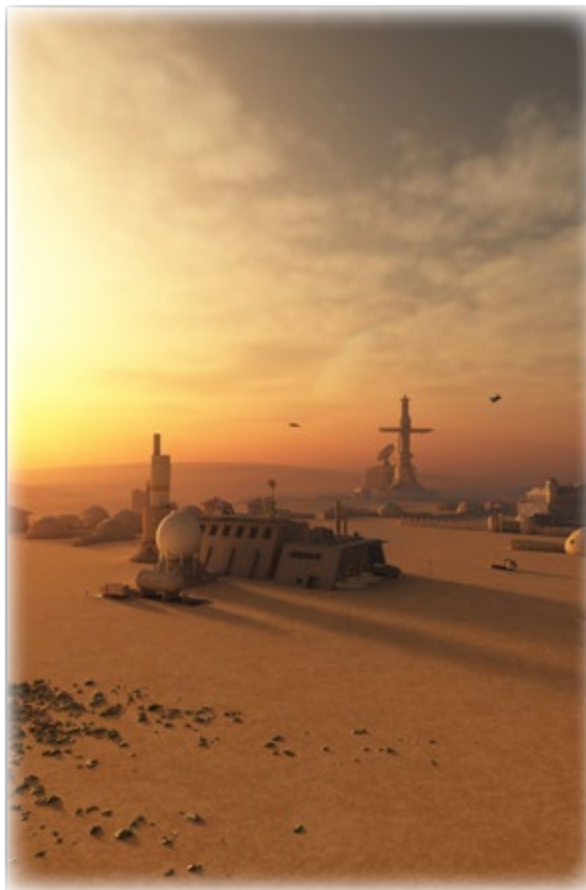
Outpost Problems and Morale

It's assumed that most outposts operate relatively smoothly. If they were a constant source of problems, most players would be disinclined to have anything to do with them, and even benign issues can eat up a lot of time at the table. As such, the GM should generally assume that a PC-run outpost isn't going to have problems unless the PCs mistreat it or the situation gets unusually dangerous.

Mistreatment of an outpost generally involves a lack of food, supplies, payroll, or necessary medical care. Introducing very antagonistic new additions to the outpost's population can also count, as can other behavior that upsets the workers.

In such cases, the GM makes a Morale check for the outpost. Most workers have a Morale of 8, though exceptionally dedicated organizations may have a higher score, or a sullen and restive labor force have a lower one. If this is a repeated provocation, penalize the check by -1 or more. On a success, the workers swallow their anger and proceed with a minimum of grumbling.

On a failure, the workers will strike, slowdown, or otherwise refuse to work effectively for 1d6 days to express their displeasure. They'll maintain any critical facilities necessary for their own survival and comfort, but extraction and processing work will certainly cease. If this is a repeated provocation and the staff have already shut the place down recently, those 1d6 days turn into 1d6 weeks. Truly angry populations might be useless for 1d6 months. An employer who pushes their staff



beyond even that can expect to see outright rebellion and seizure of the works.

Tyrannical employers can use guards, strikebreakers, or other forms of coercion to end strikes or tamp down unrest. The details of such measures will be up to the GM, but such work-camp arrangements are seldom as effective or productive as more peaceful ones.

Disruptive events will happen at the GM's discretion, usually in response to some risky choice the PCs made about how to run their outpost or acquire their resources. The earlier conflict tables in the book can be used to gin up a quick point of contention that the PCs will have to resolve if they want to keep things running smoothly.

It's worth adding a disruptive event each time the PCs push their outpost or take a risk on the world. Otherwise they can be added once every so often, when the GM needs a bit of quick filler adventure or wants to remind the PCs of the issues inherent in running an outpost. The key is not to make the place such an annoyance that it distracts the players from their other intended aims. Even so, it shouldn't be a costless, effortless well of income either, and throwing something at the party every so is justifiable.



PLAYER OUTPOST SUMMARY

This page serves as a printable quick reference for those who need to run a PC outpost's books between adventures. NPC outposts can generally be handwaved, as their economic situation may be much more complicated.

Pay Expenses

If food and supplies aren't locally produced, they'll need to be shipped in. One ton of cargo space can hold 5,000 credits worth of food or outpost supplies.

Buy Food: 5 credits/day/person who isn't being fed by a local Micro-Greenhouse or other production.

Buy Supplies: 5 credits/day/person who isn't being supplied by a local Industrial Processor with suitable inputs.

Make Payroll: 30 credits/day/staff member. Unemployed population is not paid directly by the outpost's owner.

Maintenance: 5% of total facility cost/year, which may be covered by a dedicated local Industrial Processor's output.

Maintenance Lack: If the maintenance fee isn't paid, there's a 1 in 6 chance each month of a facility breaking down and adding 10% of its cost to the maintenance debt.

Gain Income

If the outpost has staffed Extraction Industries or Industrial Processors the goods can be packed up for sale to offworld traders, assuming the outpost is visited by them.

Extraction Income: Each Extraction Industry staffer produces 10/50/100/200 credits worth of goods, depending on what materials they're extracting. A week's work averages out to seven times this amount.

Manufacturing Income: Each Industrial Processor staffer turns so many credits worth of raw materials into 200% of their value in finished goods. Each worker can make up to 500 credits a day worth of products. A week's work averages out to seven times this amount.

Shipping Goods Out: Sell the goods to traders or stockpile them. One ton of raw materials is worth 300/1,500/3,000/6,000 credits depending on the base value of the material. One ton of finished goods is worth twice their raw material inputs.

Check for Worker Injury

If the outpost is in a dangerous environment or is undertaking risky work, there's a monthly chance that someone will be seriously injured and need a Medical Clinic, ship's medbay, or other care too serious to be handled by a medkit.

Check for Injuries: If working in a reasonably dangerous environment or task, roll 1d6 each month; that percentage of the workforce is seriously injured, rounding up partial workers. If a Clinic or Hospital isn't available to take them, they've a 50% chance of dying. Safer or more dangerous situations may use different die sizes or modifiers.

NAME:		OUTPOST RECORD		
LOCATION:				
PATRON:				
INCOME/DAY:				
EXPENSES/DAY:		POPULATION:		
TOTAL VALUE:				
FACILITY	POP	STAFF	PWR	WRHSPC
TOTALS:				
NOTES:				

In Case Of Privation

If the outpost hasn't got enough food, supplies, pay, or medical care, unfortunate things will start to happen.

No Food: A person is too weak to work in three days, dead in two to four weeks.

No Supplies: Citizens will become angry and may rebel or refuse to work.

No Pay: Staffers won't work unless it's to preserve their lives.

No Medical: Staffers may strike or slow down if people die on the job due to lack of medical facilities.

No Housing: Death, if the world is uninhabitable, or severe displeasure otherwise.

Angry Locals: When locals are angered, make a Morale check, at a penalty if the provocation is repeated or severe. On a failure, they'll work only critical jobs for 1d6 days. Repeated annoyances turn days into weeks, then months, assuming outright rebellion doesn't happen first.

AN EXAMPLE COLONY

On the remote world of Cruzado our intrepid band of heroes has successfully defended a local lostworlder tribe from the depredations of pirate slavers. In gratitude, the locals show the PCs to the lode of strange yellow ore that the pirates were forcing them to work. A scientist among the PCs recognizes it as a very pure strike of aurite, a mineral that can be processed into valuable high-efficiency circuitry. While the natives have no intention of working in the dangerous mines, they're willing to let the PCs set up a mining outpost there in exchange for their protection against other nefarious offworlders.

The first thing the PCs do is estimate the size and accessibility of the aurite vein. The GM tells them that it's big enough to support about fifty workers for the foreseeable future. This means that two fully-staffed Minor Extraction Industry facilities should be able to keep active on the strike.

The GM has decided that the aurite is a Valuable commodity, so each worker would be able to dig 100 credits worth of it each day. At 50 workers, that's 5,000 credits worth of ore a day, or 25,000 a week. That's a healthy sum of profit, but the PCs need fifty capable workers on Cruzado to do it, and the natives refuse to get near the mine.

The PCs pause to do some calculating, and come to the conclusion that they can't afford to run two Extraction Industry facilities. It'd take too much Workspace and Power to keep two facilities running, plus the Residences, Medical Clinic, and Starport the outpost would need to handle traders. Reluctantly, they scale back to a single Extraction Industry for now.

Ultimately, they decide to build one Basic Residence, since Cruzado is a human-habitable world. To this, they add a Utility Complex to hold the Medical Clinic and a pair of Small Fusion Plants. A Small Work Facility houses their Minor Extraction Industry for digging out the ore, and they crown it with a Minimal Starport that can at least handle refueling for the frigate-class vessels that most far merchants pilot.

They also need to recruit offworlders willing to work on Cruzado. Fortunately, this isn't the first community the party has saved from slavers, and they can recruit fairly easily from their old allies. They'll still need to pay transport fees there and the price of a return ticket, which the GM decides amounts to 2,000 credits per worker after consulting the passage prices on page 77 of the rulebook.

In addition, the PCs need to think about colony defenses. The natives are friendly, but pirates have hit the world before. The party decides that a single Stinger Battery should be enough to discourage any pirate frigate from landing nearby.

The workers will need food, and the party debates about having it shipped in from offworld. Rather than deal with the risk of an external supply line, they persuade the natives to sell food to the outpost in exchange for off-worlder goods at the usual price rate.

The same amount of general supplies are also needed, and without a local production center or a native population able to provide them, the PCs reluctantly pencil in the need to have a ship drop off a regular supply. Perhaps a shipping agreement can be made with aurite buyers.

The players pause to total everything up, and see the new outpost will cost 550,000 credits, with a yearly net income of 305,000 credits if they do nothing special to sell its output. The construction materials will also require 465 tons of cargo space, and actually erecting all the facilities will need 235 man-days of labor from the staff.

Rather than spend three months shuttling cargo in their free trader, the PCs decide to find a freighter captain with some extra space. They also decide it's a good time to call on an old merchant acquaintance of theirs, and see just what favors he needs to get them a better aurite price.

NAME: ERIC'S DEMISE	OUTPOST RECORD
LOCATION: CRUZADO	
PATRON: THE HEROIC ADVENTURERS	
INCOME/DAY: 2,500 CR	
EXPENSES/DAY: 1,600 CR	POPULATION: 40
TOTAL VALUE: 470,000 CR, WITH 23,500/YEAR MAINT.	

FACILITY	POP	STAFF	PWR	WRHSPC
BASIC RESIDENCE	100	0	-2	0
SM. FUSION PLANT (2)	0	2	16	-6
UTILITY COMPLEX	0	5	-2	10
EXTR. INDUSTRY, MINOR	0	25	-2	-5
MEDICAL CLINIC	0	3	-2	-3
SM. WORK FACILITY	0	1	-1	5
STARPORT, MINIMAL	0	3	-4	0
STINGER BATTERY	0	1	-1	0
TOTALS:	100	40	2	1

NOTES: FOOD IS BOUGHT FROM LOCALS. SUPPLIES NEED TO BE SHIPPED IN AT 5,000 CR/TON. IF NOTHING GOES WRONG AND TRADERS KEEP COMING, WEEKLY NET INCOME IS 6,300 CR, WITH 23,500 CR IN MAINTENANCE DUE ONCE PER YEAR.



INFRASTRUCTURE FACILITIES

These facilities generally revolve around basic requirements of shelter, power, food, and transport accessibility. Their particular appearance or details can be determined by the buyer, and some worlds might manage to produce more or less efficient models.

Comm Center: While expensive human-portable FTL comms do exist for in-system communication, and it's possible for a skilled technician to use standard radios for orbital or more distant radio communication, a comm center is specially designed to handle planetary, orbital, and interplanetary communications needs. A technician is always on duty to monitor the sensors, and has a chance to detect in-system starships as normal. The included dozen basic comm satellites allow small-scale planetary communication, provided a ship can put them in orbit. Internal outpost comm lines are often routed through the comm center.

Environmental Generator, Large: A large-scale version of its smaller cousin, this environmental generator requires several hectares of space in order to fan out its array of mirrors, windmills, geothermal taps, and other power units. It requires five dedicated techs to keep it in proper operation condition, or else its effectiveness is halved.

Environmental Generator, Small: A versatile powerplant that includes wind, solar, thermal, and hydropower components, this generator can produce a small amount of power from the natural environment of most planets. The tech is somewhat fragile due to its complexity, however, and requires at least one tech to maintain and calibrate it; without such attention, its power generation is halved.

Fusion Plant, Large: A large municipal power plant, this fusion plant runs on water or hydrogen. If those substances have to be shipped in to fuel it, a frigate-load of fuel is enough to keep the plant running for a week, or longer if the output is lowered proportionally.

Fusion Plant, Small: Simple, reliable, and effective, the fusion plant is the typical favorite power plant of planetary outposts. It requires only a steady supply of liquid water or hydrogen to function, and those substances are usually common on any human-inhabited world. Outposts on exceptionally hostile planets or barren asteroids may have to ferry in hydrogen for the plant, however; a single frigate-load of fuel such as can be gathered with the Fuel Scoops fitting is enough to operate it for a month.

Gravitic Generator: A precious pretech generator that somehow uses the gravitic field of any celestial body larger than an asteroid to generate seemingly unending amounts of electrical power. Once the gravitic generator is installed and calibrated to the planet's gravity field, it will continue to emit power until it is deactivated or damaged beyond repair. These units are rare in the present day, but they were some of the most common generators

supplied to those Second Wave colony efforts that were supported by the Terran Mandate, and surviving examples might be found in long-lost ruins.

Hab Block, Barracks: Grimly efficient, the barracks block packs its inhabitants into shared dormitories, general mess halls, and group lavatories. Inhabitants have no more privacy than a curtain and a personal locker can grant them, and their power allowance is barely enough to charge a Type A cell daily. Minimal provision is made for families, and these blocks are usually reserved for single laborers, prisoners, or similar low-status workers.

Hab Block, Basic: While spacious compared to the cramped quarters of a starship's sleeping berth, this hab block is not particularly generous to its inhabitants. Each has a small apartment, with shared hygiene units and minimal recreational or work space. Power demands are relatively minimal, however, and the hab block construction is very sturdy, with walls equivalent to reinforced concrete.

Hab Block, Forward Shelter: A simple, sturdy structure designed to assemble itself from foamed lightweight composites and pressure-sculpted ceraplasts, the forward shelter is a single pressurized building with about 500 square meters of floor space. As a structure built to function even in the absence of a proper power generation plant, a Type B power cell can keep it running for one day, albeit with considerably less comfort than usual. It has living space for up to 10 friendly inhabitants, and a small amount of space designed to accommodate industrial or mechanical work. It can be erected by a single person in six hours, but once built it cannot be practically moved.

Hab Block, Luxury: Externally similar in size and appearance to its more basic cousin, this hab block is appointed with spacious suites, private baths, graceful recreational spaces, and a considerably more generous power budget than is allotted to less blessed inhabitants. It's not unusual for the elite of a given outpost to be allotted these spaces.

Life Support, Automated Ranch: An integrated unit that grows both livestock and the fodder they need for survival, the automated ranch can provide a steady supply of protein-rich food to outpost inhabitants, often of a very high quality compared to that of poorer TL4 worlds. The livestock require a planet with tolerable atmospheric and temperature conditions, however, and there is a risk that local predators may find them to be digestible. At least twenty hectares of open space is needed for pasturage.

Life Support, Hydroponic Garden: This hydroponic array has room not only for plants, but a small selection of livestock, greatly increasing the variety of foodstuffs that can be produced. The additional size improves economies of scale in production as well, and helps the unit serve as a water recycling plant for up to 100 users. The garden requires at least a month to spin up production when first established or when recovering from collapse.

Facility	Cost	Tonnage	Pop	Staff	Power	Workspace
Hab Block, Forward Shelter	15,000	10	10	0	-1	1
Hab Block, Basic	50,000	100	50	10	-4	1
Hab Block, Luxury	50,000	150	25	5	-4	1
Hab Block, Barracks	25,000	100	125	10	-6	1
Residences, Basic	25,000	50	50	0	-2	0
Residences, Luxury	50,000	50	25	0	-4	0
Quantum House	N/A	1	50	0	0	2
Life Support, Micro-Greenhouse	10,000	10	Feeds 10	3	-1	-1
Life Support, Hydroponic Garden	100,000	50	Feeds 100	10	-5	-5
Life Support, Robotic Farm	200,000	50	Feeds 500	5	-5	-2
Life Support, Automated Ranch	250,000	100	Feeds 250	10	-2	-3
Comm Center	100,000	10	0	3	-5	-1
Environmental Generator, Small	50,000	10	0	1	4	0
Environmental Generator, Large	150,000	50	0	5	20	0
Fusion Plant, Small	40,000	30	0	1	8	-3
Fusion Plant, Large	100,000	100	0	3	20	-10
Gravitic Generator	N/A	5	0	0	40	-1
Medical, Clinic	50,000	25	0	3	-2	-3
Medical, Hospital	5,000,000	100	0	30	-10	-10
Starport, Minimal	25,000	25	0	3	-4	0
Starport, Basic	500,000	100	0	15	-10	0
Starport, Advanced	10,000,000	250	0	50	-25	0
Starport, Military	50,000,000	1,000	0	120	-50	0
Utility Complex	50,000	50	0	5	-2	10
Utility Shed	10,000	10	0	0	0	2

Life Support, Micro-Greenhouse: A small greenhouse designed to produce a nutritionally-balanced supply of plant-based and synthesized foods using preloaded seed stocks and fabrication templates. It also has a sewage processing unit capable of recycling water for up to 10 inhabitants. While some additional water will need to be added to account for losses eventually, this is normally only a problem if the outpost is completely deprived of water or the oxygen and hydrogen needed to synthesize it. The micro-greenhouse requires a month of growth before it can start producing food; the same delay is needed if its crops are wiped out by some disaster or disease.

Life Support, Robotic Farm: This infrastructure unit requires a habitable world, with a breathable atmosphere and human-tolerable temperature ranges. By taking advantage of outdoor plant growth and an elaborate array of robotic tenders and soil monitors, a farm can produce large amounts of staple vegetables and cereals with minimal human oversight. Such farms provide all normally-required irrigation and temperature controls, but they remain vulnerable to extreme weather conditions, and require at least five hectares of open space.

Medical, Clinic: A basic but serviceable clinic designed to provide both common health services and emergency medical treatment for serious cases. While it lacks the resources for the regrowth of lost limbs, it can provide somewhat crude cybernetic prosthetics. Significant diagnostic tech is available for identifying new diseases and environmental hazards, however, and the clinic can formulate limited pharmaceutical treatments for such dangers. A clinic has room to handle up to 10 serious medical cases at once.

Medical, Hospital: A small but full-fledged TL4 hospital, one capable of cloning lost limbs and providing the full range of modern TL4 medical treatments that would be normal on a civilized world. The hospital is equipped with advanced diagnostic and drug-synthesis tech, and can provide solutions to much more complex problems than a clinic could manage. It's also capable of synthesizing effectively unlimited amounts of tailored anti-allergenic drugs such as might be needed on some worlds, though these synthesized anti-allergens are rarely stable enough to stockpile large amounts of them in advance. A hospital can handle any amount of medical care that might reasonably be needed in an outpost.



Quantum House: A strange example of latter pretech expeditionary gear, a quantum house appears to be nothing more than a large archway, broad enough to drive a vehicle through. When fixed on a celestial body no smaller than the Earth's moon, however, it can be activated to create an interstitial dimensional space beyond the gateway, one that manifests as a very comfortable, well-kept habitat block with room for fifty inhabits, or twice as many if they crowd. Vehicles no larger than a freight truck can pass through the archway and into a receiving area, while the space inside appears to be self-powered and to remain at a comfortable atmospheric and temperature state. There's even an integrated eco-recycler system that provides all necessary food and water for fifty inhabitants. Once established, the quantum house cannot be moved until it is deactivated from inside the pocket space; once turned off, any object or person left within the space reappears outside. The same effect occurs if the archway is damaged beyond repair, which will happen if it suffers 100 points of damage from heavy weapons.

Residences, Basic: Feasible only on planets with breathable atmospheres and human-tolerable temperatures, these single-family dwellings are relatively luxurious compared to standard hab block housing. While they're equipped with normal heating and cooling capacities, they aren't sealed against the local atmosphere and require much more space than a dense apartment block. They can still be used as de minimis shelter in the absence of a power plant, but they're much more uncomfortable in such a circumstance.

Residences, Luxury: Just as basic residences, these single-family homes require a tolerable atmosphere and climate. Even so, they're roomy, graceful, and well-equipped with all the conveniences that might be expected by an affluent TL4 family. Such housing would usually be something reserved for the elites of most starfaring worlds.

Starport, Advanced: The starport is the equivalent of some secondary TL4 planetary starports, and is unusually sophisticated for an outpost port. It has all the fittings of the basic starport, but can refresh, refuel, and resupply up to a cruiser-class vessel if shuttles are available to ferry the goods into orbit. The repair facilities can fix even destroyed fittings on frigate-class or smaller ships, and all hull damage can be repaired as well. Sophisticated quarantine and movement control barriers are in place, ensuring that cargo inspections and crew examination can be performed before visitors are allowed into the outpost.

Starport, Basic: The facilities here are simple, but good by the standards of a planetary outpost. The control tower has a full sensor array capable of detecting ships in the system as per the usual rules, and communications gear equivalent to that of the Comm Center infrastructure unit. There are enough repair facilities here to allow for standard ship maintenance and the repair of damaged but

not destroyed ship systems. The fuel bunkers can hold up to ten frigate-loads of fuel and process a load a day from available hydrogen or water. Assuming the outpost is not strapped for food, it can also refresh the life support and ship supplies of frigate-class ships or smaller. Basic walls and gates prevent a planet-side starport from being easily entered or left by intruders.

Starport, Military: A starport of this grade is usually found only on naval outposts, and is almost always built into asteroids, small planetoids, or other deep-space microgravity locations. It has all the features of an advanced starport, but can provide life support, refueling, and supply service to capital-class ships, along with munitions replenishment, military tech maintenance, and the handling of encrypted high-security reports and files. It's a hardened facility with strict ingress and egress requirements and forty heavily-armed security personnel on duty at all times. If necessary, it can isolate itself from the outpost to serve as a fortified redoubt, its systems and backup batteries adequate to support its staff and any visiting crew for up to a month.

Starport, Minimal: This starport is little more than a flat landing zone, a control hut, and a hydrogen pump. The control hut is equipped with comm tech adequate for communicating with ships in orbit, but no longer-range sensors are provided. The hydrogen stores are sufficient to provide up to five frigate-sized loads of fuel; assuming water or atmospheric hydrogen is available, the pumps can process it into usable fuel at the rate of one load per day. The starport does not have the equipment to allow for ship maintenance here, nor do they have any significant amount of repair supplies. They can flush a ship's life support system to refresh it, but onboard supplies are limited to whatever the outpost might have warehoused here. There is very little in the way of quarantine control if this infrastructure is established on a planet, little more than a polite request that the ship crew remain inside until a med tech can look them over and an inspector examine their cargo.

Utility Complex: A compact, secure building designed to house colony infrastructure units such as hydroponic gardens, power plants, medical facilities, or the like. Its workspace can be used for any infrastructure facility, but both it and the smaller Utility Shed are not designed to support industrial use, and such facilities cannot be sited here.

Utility Shed: A small structure meant to house a single colony facility, a utility shed requires minimal janitorial or technical maintenance. Because it lacks an easily-integrated architectural format, however, any facility meant to be housed in a utility shed must fit completely in a single shed. As such, a Comm Center or Robotic Farm could operate out of a shed, but a large Fusion Plant or a Medical Clinic would be too big.

INDUSTRIAL FACILITIES

These facilities produce valuable outputs for the outpost owners. Keep in mind, however, that every outpost site has a limit to the extent of its exploitation. It's for the GM to determine how many workers can actually effectively work a site.

Bot Maintenance Bay: The robotic equivalent of a Medical Clinic, this workspace is capable of repairing and maintaining damaged robots of all kinds. Up to ten wrecked or seriously damaged robots can be rebuilt each month, and ordinary reprogramming or modification can be conducted with the available tools and parts.

Extraction Industry, Major: This unit functions much as its smaller equivalent does, albeit it has room for a larger number of workers to effectively extract the resource. It's up to the GM to decide how much of the resource exists, and how much total wealth can be pulled from it before the lode runs out.

Extraction Industry, Minor: Whether miners, foresters, crystal-cutters, atmosphere reproducers, or some other extractors, this facility is designed to pull valuable raw materials from the area. The precise worth of these materials will depend on the GM's judgment. Depending on the economics, these raw materials might be packed up as-is for shipment offworld, or they might be further processed in the colony by the Industrial Processor facility.

Industrial Processor, Major: This large-scale processor unit works much as its smaller equivalent does, but allows for far more workers to process the raw materials, each producing 500 credits a day of finished goods.

Industrial Processor, Minor: This unit can take the raw materials produced by an Extraction Industry and fabricate them into finished goods of a particular type. The processor might make spare parts, or starship repair materials, or laser rifles, or anything else that it might plausibly have the production plans for. The raw materials must be appropriate for creating the goods; piles of ores and minerals might be fine for building laser rifles, but they won't work as feedstock for luxury perfumes. A given amount of raw materials can create 150% as many finished goods. Thus, if the outpost has 3000 credits worth

of ores in its storage units, it can build 4500 credits worth of laser rifles. Each worker can fabricate up to 500 credits worth of goods each day.

Labor Bot: This is no more than a standard industrial work bot as given on page 197 of the core book, one programmed for outpost labor duties. They're too simple-minded to function without human or VI oversight, but they can help erect outpost facilities or do simple manual labor when properly monitored. It's possible to employ them in Extraction Industries or Industrial Processors, but the small scale of outpost production makes it much more difficult to use them efficiently; a bot can replace a human worker but they are only one-quarter as efficient in output, and they require additional human bot wranglers. A single human bot operator can usually oversee up to five labor bots at a time, and a functioning outpost power grid or operational starship can power any reasonable number of them.

Vehicle Shop: Any outpost can manage a shelter for a gravtruck and perform necessary maintenance and most needed repairs with hand tools and patience. Outposts that rely on large numbers of vehicles or that need to overhaul a totaled piece of equipment may need to establish a vehicle shop. The shop can rebuild even wrecked vehicles at half of their original purchase price. Using spare parts and fabricated components, TL4 vehicles can also be built here at 75% of their list price, with motorcycle-sized ones built in a day, car-sized ones built in a week, and truck or larger vehicles fabricated in two weeks. The shop is sufficient to build or install vehicle modifications as given in the *Engines of Babylon* supplement.

Work Facility, Large: A large work complex with space for a substantial factory or significant resource extraction site. The industrial-grade power conduits, ventilation facilities, and safety barriers allow work to be performed here that might be too dangerous for an ordinary workspace.

Work Facility, Small: This structure is a smaller and less efficient space than its larger version, but requires minimal power and maintenance while providing the necessary infrastructure for heavy industrial labor.

Facility	Cost	Tonnage	Pop	Staff	Power	Workspace
Bot Maintenance Bay	50,000	25	0	3	-4	-2
Extraction Industry, Major	500,000	500	0	Up to 250	-15	-20
Extraction Industry, Minor	100,000	100	0	Up to 25	-2	-5
Industrial Processor, Major	1,000,000	250	0	Up to 100	-20	-20
Industrial Processor, Minor	200,000	50	0	Up to 10	-2	-5
Labor Bot	2,000	1	0	0	0	0
Vehicle Shop	50,000	25	0	5	-5	-5
Work Facility, Large	250,000	200	0	4	-4	25
Work Facility, Small	100,000	50	0	1	-1	5



DEFENSE FACILITIES

Even on an airless rock, an outpost still runs the risk of hostile attention from pirates or enemy worlds. On inhabited planets, native powers often have an acquisitive attitude towards the wealth of offworlders. In both cases, a healthy set of defenses can do much to ensure an outpost's peaceful existence.

These facilities provide various built-in defenses and additional military resources for the outpost. Many of them require dedicated staffers to keep them operating or to monitor their sensors.

For facilities such as the Pacification Force Barracks or the Armored Patrol, the listed staffing requirements include only operating and support staffers. Owners who want to have soldiers for the barracks or troopers for the APC will need to hire them separately, usually at 30 credits/day if not more.

The guns and defensive systems here generally stockpile battery energy and munitions during lulls. The listed Power costs will keep them operating normally during ordinary battles or operation, but extended use may exhaust their capacitors or empty their magazines at an inopportune time.

Defense Facility Attack Rolls

If a target is attacked by a defense facility, such as an infiltrator targeted by a Fauna Control System, its expert system makes an attack roll with a base +2 bonus to hit. If a human gunner is operating the weapon, they make a normal Dex/Shoot or Int/Shoot attack roll.

Defense facility weapons generally have a range of up to a kilometer, with far more in the case of Stinger Batteries or other anti-air defenses.

Skulking past a defense facility's sensors usually requires a Dex/Sneak skill check against difficulty 10 or more. If a human operator is actively monitoring the system, the infiltrator's roll must also beat an opposed Wis/Notice check on the operator's part.

Antipersonnel Defenses: Various small lasers or projectile weapon turrets are mounted around the perimeter of the outpost, all of them perfectly capable of killing any terrestrial animal and most terrestrial vehicles. The number of individual units that can be brought to bear on a single target will vary depending on the layout of the outpost, but usually two of them can fix any single target. On a hit, the target takes 2d6 Heavy damage.

Armored Patrol: A gravitic armored personnel carrier as described in *Engines of Babylon* is provided by this facility, along with its pilot, gunner, and maintenance crew. The carrier is immune to non-Heavy weapons, is equipped with an anti-personnel laser that does 3d6 damage and can fire to suppress, and can carry up to fifteen soldiers. Most primitive forces are utterly helpless against the APC, and the grav boosters on the vehicle can overcome all but the highest walls or most difficult terrain. While the crew can perform standard maintenance on it and repair minor damage, a vehicle repair facility is needed for major repairs.

Colonial Guard Armory: This well-fortified building contains enough TL4 weaponry and armor to kit up to fifty locals along with command and control facilities to handle military communications within the outpost. The usual equipment consists of laser rifles and combat field uniforms, with moderate amounts of grenades, anti-vehicle weaponry, and portable railgun emplacements. Establishing such an armory also establishes regular training cycles for the locals. It's not enough to turn factory workers into soldiers, but it ensures that the colony can mount an organized resistance to hostile military action.

Defense Bot Force: Five soldier bots as per page 197 of the rulebook are included in this package, along with basic maintenance parts and TL4 weaponry. These bots are most often used on uninhabited planets with dangerous fauna; their expert systems have difficulty dealing with intelligent opponents without a human bot wrangler to guide them, but they can tirelessly patrol an area for hostile alien life.

Fauna Control System: One of a number of less-lethal defensive turrets designed to be set up around the outpost at various points along the perimeter. They use sound waves, stench weaponry, microwave stinging, or other measures to drive away unwanted native fauna without killing the creatures. If they target an enemy and hit, the victim takes 2d6 points of disorienting damage. If such a victim reaches 0 hit points, they regain 1 and must flee the area.

Hardened Perimeter: A physical wall of reinforced plasticrete or similar fortified material can be built around the entire outpost. This wall is usually sufficient to deter all but the most determined native predators and can defeat local humans who haven't preserved a tradition of siegecraft. The cost of such a wall depends on the size of the outpost, usually amounting to 5% of the total outpost cost aside from itself. If the outpost grows later, this facility must be updated with the difference in cost.

Micro Braker Gun: A scaled-down version of more conventional civil-defense braker guns, this facility constantly scans for incoming meteorites, crashing vehicles, dropped munitions, or other orbital hazards. When such an object is calculated to impact near the outpost, the Micro-Braker gun uses advanced gravitic projection to nudge it towards a safely-uninhabited area. Outposts equipped with these guns cannot be successfully bombarded by anything short of a capital-class ship's weapons or a fleet of similar mass.

Pacification Force Barracks: More a small fortress than a conventional colony building, this barracks is typical of a military garrison sent to suppress troublesome natives or ensure the cooperation of a local population center. It can house a staff of 100 trained garrison soldiers, supporting them with as many trained logistical staffers, and has extensive security measures to prevent unauthorized access. Each soldier employed here is equipped with assault

Facility	Cost	Tonnage	Pop	Staff	Power	Workspace
Antipersonnel Defenses	50,000	25	0	3	-4	0
Armored Patrol	300,000	25	0	5	-5	-5
Colonial Guard Armory	100,000	20	0	0	-1	-2
Defense Bot Force	60,000	10	0	1	-1	-1
Fauna Control System	25,000	10	0	0	-2	0
Hardened Perimeter	Special	50	0	0	-2	0
Micro Braker Gun	10,000	5	0	1	-1	0
Pacification Force Barracks	2,000,000	200	200	100	-10	5
Quantum ECM Projector	25,000	5	0	1	-2	-1
Sealed Perimeter	Special	250	0	5	-10	0
Smart Minefield	Special	25	0	3	-2	0
Stinger Battery	15,000	5	0	1	-1	0

armor and a laser rifle, and the armory includes a wide range of military-grade hardware. If cut off from the rest of the outpost, its reserves can keep it and its occupants functioning for up to three months.

Quantum ECM Projector: Outposts at risk of attack from TL4 forces often find it necessary to install quantum ECM in order to prevent drone attacks or murderously accurate guided munitions. This projector is capable of scrambling remote control within a 50 kilometer radius. As with all quantum ECM the effect is indiscriminate when it is operating.

Sealed Perimeter: This structure physically seals the outpost off from the world around it. A temperate, breathable atmosphere can be maintained within the dome, energy field, or bunker, provided the perimeter generators keep functioning and Power keeps flowing. This perimeter is extremely difficult to penetrate from either side, and usually requires would-be infiltrators to have TL4 equipment to bypass its many monitors and sensors, or else use military-grade explosives to cause a breach. It's also extremely expensive to build, usually amounting to 20% of the total outpost facility cost. If the outpost grows later, this facility must be updated with the difference in cost.

Smart Minefield: The perimeter of the outpost is fortified with a wide band of smart mines, ones that can be

activated or deactivated remotely by the defenders. The sensitivity of the minefield can be adjusted to apply only to vehicles or large native animals, or dialed down to trigger on human-sized disturbances. A human has a 25% chance of making it through the field without incurring 2d6 Heavy damage from a mine, or a 50% chance if they follow in the footsteps of a successful comrade. Detecting and removing these mines is generally not feasible without TL4 tools. The package comes with enough spare mines to maintain the field's ordinary operation, and costs 10% of the outpost's total facility cost.

Stinger Battery: Cheap, simple, and reliable, the exact nature of a Stinger Battery varies from world to world. Some rely on large arrays of laser tubes while others prefer physical munitions. The most advanced component of any Stinger Battery is its radar, which is programmed to identify and lock on to ships or shuttles that are within the outpost's line of sight. It can attack as usual for a defensive facility, and does 3d4 Heavy damage on a hit. Against starships or vehicles, it counts as a starship-class weapon and so applies its damage normally. It can hit any aerial target within the facility's field of view, but it cannot be depressed far enough to engage ground targets. Stinger Batteries usually serve as an outpost's first line of defense against pirate raids, but they're also a threat to any ship that dares take off from the outpost without permission.



DEEP SPACE OUTPOSTS

Most outposts are planted on a major planetary body, if only for the fact that that's where most of the rarest and most valuable materials in a star system are found. The metals and raw materials found in the average asteroid belt aren't usually enough to entice a patron into building a full-fledged industrial facility there.

At times, however, a group finds it necessary to build their refuge far from a gravity well. There may be some rare mineral found in a particular asteroid belt, or scientific research might need an outpost planted far from the system's primary, or the inhabitants might just have a profound aversion to visitors. For whatever the reason, these people need to construct an outpost deep in the black.

While it's possible to haul an entire space station to the site, using the statistics and rules in the core rulebook, such a solution isn't always the best or cheapest for housing large numbers of crew. More often, the colonists use ship engines and nudges to push a suitable asteroid into an acceptable orbit at whatever distance they require, and then build their outpost into the rock.

Asteroid Thrusters: These maneuver thrusters allow for the slow repositioning of the outpost elsewhere in the system. It takes up to a month to change system regions this way, but the thrusters have such a limited output that they don't show up well on energy sensors. The movement has no chance of being noticed on system scanners unless a searcher is specifically looking for the outpost.

Black Body Shielding: The surface of an asteroid can be mantled in various high-tech composites that absorb almost all conventional electromagnetic scanning wavelengths. With the addition of thermal dump cells within the asteroid mass itself, the outpost can become almost entirely non-emissive, becoming virtually impossible to detect with conventional TL4 scanners so long as it avoids self-powered movement, radio transmissions, or

ship traffic. If a sensor scan of the outpost's region is run during a day in which one of those things has happened, the sensor scan roll has a chance of discovering its presence, albeit with a +2 difficulty modifier on the roll. If the outpost is located in deep space beyond the system rim, it can be almost impossible to find. The cost of this shielding varies with the size of the outpost, amounting to 30% of its total facility cost.

Cargo Hold: The absence of gravity and the constant bombardment of stellar radiation makes the surface of most asteroids unsuitable for storage. This facility makes a pressurized, heated, properly-maintained space of 100 cargo tons within the asteroid, suitable for holding mine outputs, finished goods, or supplies.

Cargo Umbilicals: Cheaper and simpler than a proper ship bay, these umbilicals allow pressurized transit between the outpost and an adjacent ship. Cargo transit is slow and awkward, however, amounting to no more than 50 tons a day, and a ship can't refuel through it.

Concealed Structures: The outpost is built so as to be indistinguishable from an ordinary asteroid to visual inspection and superficial scans. Any attempt to identify the outpost requires that the searcher be specifically looking for a masked asteroid in the region and take a +2 difficulty penalty on any scans. Such concealment increases the cost of all facilities by 20%, however, and every facility must be concealed if the subterfuge is to hold.

Localized Gravity Field: While it's possible for some adapted humans to survive for long periods in zero gravity, most workers will require something better than micro-gravity in their living quarters for the sake of long-term physical and mental health. This gravity field is localized around the more important parts of the outpost. It's not able to generate more than 1G, but its operators can selectively disable it when needed.

Facility	Cost	Tonnage	Pop	Staff	Power	Workspace
Asteroid Thrusters	50,000	50	0	0	-2	0
Black Body Shielding	Special	200	0	3	-5	0
Cargo Hold	10,000	0	0	1	-1	0
Cargo Umbilicals	50,000	25	0	1	-2	-1
Concealed Structures	Special	0	0	0	0	0
Localized Gravity Field	200,000	50	0	5	-5	-1
Ship Bay, Large	1,000,000	300	0	12	-10	-5
Ship Bay, Small	250,000	100	0	3	-3	-2
Ship Maintenance Facility	500,000	150	0	12	-5	-5
Staff Control Measures	200,000	100	0	9	-4	-2
Thermal Mass Buffer	250,000	150	0	6	-5	-3
Rotational Gravity	50,000	25	0	3	-2	0
Anti-Intrusion Network	Special	100	0	3	-3	-2



Ship Bay, Large: This pressurized ship bay can hold up to a half-dozen frigate-class ships or four times as many shuttle-sized ones, and provides all the functionality of a cargo umbilical when handling a larger ship. A fuel bunker can store up to six frigate-sized loads of fuel or four times as many shuttle-sized ones. This bunker can be replenished by asteroid ice or hydrogen from a suitable gas giant or star. Outposts without a ship bay or a cargo umbilical will have an extremely difficult time moving cargo and personnel to and from the outpost.

Ship Bay, Small: A smaller, less efficient version of its larger cousin, this bay can hold up to one frigate-class ship or four shuttles, and has fuel bunkers capable of holding one frigate or four shuttle-sized loads of fuel. It also functions as a cargo umbilical when dealing with cruisers or larger craft incapable of docking.

Ship Maintenance Facility: Granted sufficient spare parts, this well-equipped ship service bay can fully repair damaged or destroyed ship components on cruiser-sized vessels or smaller, and can perform annual maintenance as needed. The repair facilities can fix up to 10 hit points/day of ship hull damage.

Staff Control Measures: There's no prison like an asteroid, and this facility provides jailers with all the tools necessary to keep their workers in line. A vast array of monitoring sensors is paired with security checkpoints, sealed bulkheads, and selective atmosphere control to keep the outpost's inhabitants completely under the control of its overseers. The facilities of the outpost are hardened against casual sabotage and the tools and other equipment are designed to be disabled or destroyed should they be used in an attempted rebellion. Staffers under the control of these measures can be forced to work regardless of their wishes, though the die for monthly worker injuries is never less than 1d8 due to the tensions of the place.

Thermal Mass Buffer: For installations that really, really dislike visitors, an advanced thermal buffer can be installed in an asteroid of the appropriate composition, one allowing the outpost to survive in close proximity to a star or other hot body. This buffer requires at least a hundred tons of water or similar buffer fluid to be consumed on a weekly basis in order to vent the heat, a material requirement that is usually fulfilled by mining ships. Tracking such ships is usually the only way to locate such an outpost, as it is otherwise invisible to TL4 scanners due to the blinding energy emissions around it.

Rotational Gravity: A much simpler and cheaper alternative to a Localized Gravity Field, the entire asteroid is spun at a speed sufficient to give its hollowed interior approximately 1G of gravity-equivalent. A relatively complex ballasting system is necessary to account for mass shifts within the asteroid, but the engineering is still simpler than that of an LGF facility. Ship bays and other egress points are usually placed in zero-G at the central axis of the asteroid's spin. Unlike an LGF, individual gravitic zones cannot be controlled in such an outpost, and serious damage to the ballast system can run the risk of grave facility damage as the asteroid "wobbles" unevenly.

Anti-Intrusion Network: While asteroids are cheap and easy to find, any of them large enough to serve adequately as an outpost are also large enough to be difficult to defend from unwanted surface landings. Even if the outpost is equipped with defensive measures, there are usually blind spots or dark sides that a hostile assailant can land boarders on, if there's no space cover available from defending starships. This network mobilizes the outpost's defensive facilities, allowing them to shift around the surface and respond to boarding attempts. Such mobilization is expensive, however, and makes each such facility so improved cost five times as much as normal. Thus, a Smart Minefield that covered the entire asteroid would end up costing 50% of the outpost's total facility cost.

COLONIAL GEAR

Most TL4 equipment is built for use on populated planets, where a repair shop and a fresh supply of parts are easily obtained. Equipment meant for use on outpost worlds rarely has that luxury, and is often built simpler and tougher than standard equipment. Such design goals sometimes result in compromised effectiveness or the need for more frequent maintenance, but it's a worthwhile tradeoff against the risk of breaking critical gear with no spare on the planet.

The equipment described below is usually available on any TL4 world, though the cybernetic augments may be rarer in places without an economy based on resource exploitation and hostile environment survival.

Biomass Generator: A foldable combination wood-stove-generator, this unit is built to allow the burning of most organic matter to fuel its power production and provide warmth. Four encumbrance points worth of wood, twigs, or the like can be made to burn for 8 hours and recharge up to three type A cells in the process.

Floater Lamp: A very small antigrav node is attached to an ultra-light spotlight and video transmitter, the whole controlled by a hand-held tablet that can receive audio-visual transmissions from the lamp. The beam's spotlight can reach up to 30 meters. The unit's range is very limited, no more than 50 meters, and it must retain line-of-sight with the controller at all times. If it exceeds this range it automatically powers down and lands. The unit can be set to hover in place or follow, but actively maneuvering it requires a Move action by the controller. It also can be piloted as a drone by a person with the Drone Control Link cyber. A type A cell powers the lamp for one hour.

Mag-chain Cutter: The heavy, two-handed equivalent of a TL4 chainsaw, the mag-driven teeth of this cutter can lop trees, clear brush, cut through walls of a lighter construction than concrete, or inflict terrible wounds on those in its way. It takes one minute of operation to cut a man-sized hole in a wall or safely down a meter-thick tree. If used as a weapon, its ungainly bulk applies a -4 penalty to hit and causes the user to always attack last in a round; however, it does 2d8 damage, Shock 3/18, and uses Strength as the modifying attribute. A type A cell powers the cutter for one scene of constant use.

Micro Comm Satellite: A cheap, none-too-sturdy commodity satellite that can be thrown out a starship airlock and find its own orbital station over a world. A satellite will function for up to a year before de-orbiting, and they're trivially easy for starships to find and destroy. A minimum of four are needed for covering a small area, while 24 of them can completely cover a world. Covered areas are provided with a GPS signal and voice comms that can be accessed by anyone with a TL4 radio; the bandwidth is limited, but usually quite sufficient for an outpost. Some worlds may have a magnetic field or weather conditions that make space-based communications useless.

Native Antitoxin: This oversized stim unit contains enough specially-formulated antivenom to treat a dozen cases of poisoning from this world's native fauna or toxic local plants. If applied by an untrained user, the victim gets a second saving throw versus the poison at a +2 bonus. If applied by someone with the Heal skill, the medic can make an Int/Heal check against difficulty 10; on a success, the toxin is neutralized, and on a failure, the victim still gets the second saving throw.

Panic Shelter: A single-use emergency shelter that can be triggered and dropped as a Main Action. The shelter forms a thin, pressurized bubble around the user, slightly wider than they are tall and sufficient for only one person, or two if they cling tightly. Within the bubble a tolerable temperature and atmosphere can be maintained for up to six hours in all but the most extreme planetary conditions. The bubble also transmits a radio distress call on all the usually-monitored frequencies, along with a location for the shelter. Any blow or cut will destroy the bubble. The shelter includes several vacc suit patches which can be applied by a user inside the bubble.

Patch Putty: This repair and jury-rigging tool comes in the form of one kilo of soft gray putty. When molded by a user and then electrically charged with the included tab, it hardens immediately into a substance as tough as steel. The putty can adhere to surrounding materials if the user cleans and preps them beforehand; otherwise it sticks only to itself. The hardened putty can be removed easily with a metatool and a minute of work. Patch putty functions best in small amounts, and objects including more than 10 kilos of the substance tend to fail to harden. Patch putty can be used only once.

Perimeter Alarm: Several small, easily-hidden sensors can be placed to ring an area up to 50 meters in diameter, creating an invisible sensor bubble around the enclosed area. Any organic creature that crosses the perimeter will trigger an alert on the unit's control pad, showing a live audiovisual feed from the nearest sensor. The sensors have good low-light vision and can be tuned to ignore creatures of a certain size or smaller. Defeating the alarm requires a skilled infiltrator and an Int/Sneak or Dex/Sneak check at difficulty 10. One type A cell powers the perimeter for 24 hours in total.

Poisonous Bait: A mass of extremely toxic matter is scented and textured to appear as something edible to the local carnivores. If thrown to them or laid out to tempt them, a hungry beast has a 50% chance of eating it before bothering with other prey. An animal who eats it must make a Physical save or take 4d6 damage, vomiting it out as a Main Action next round if the save is successful. Most bait is formulated for the fauna of a particular world and is non-toxic to humans.

Portable Ladder: Advanced composites allow a ten-meter ladder to collapse down to a fist-sized mass. Adhesion struts along the surface allow the ladder to safely affix to any solid, mostly-dry surface. It can be set up or taken down with one minute of work, and can support up to 300 kilos of weight at a time.

Porterbot: These four-legged work bots have statistics equivalent to an Industrial Work Bot as per page 197 of the rules, but have no combat ability. They're designed to navigate extremely difficult terrain and serve as a stable platform for carrying heavy loads or incapacitated humans, being capable of carrying up to 300 kilos of general cargo or 60 items of encumbrance. Their basic expert system responds to verbal commands or remote compad control, but they're too simple-minded to operate independently. They have two manipulator arms that can load and unload their own cargo, but delicate work with them is impossible. A Type B cell powers them for a week.

Survival Guide: Available on most worlds where an established colony has had a chance to scout the locale, a colonial survival guide is a simple handbook on waterproof plastic with advice, warnings, and instructions relevant to surviving the dangers of a particular planet. It includes information on local flora and fauna, native societies, weather, edible life forms, and other important topics, along with a map of the colony and its surroundings and a compass. While the information is no substitute for the Survival skill, it can prevent a reader from making very large mistakes.

Tailored Adaptation Implant: Some worlds require a constant supply of tailored anti-allergens to survive the native pollens, or have native life with indigestible tissues, or are rife with an endemic plague. This simple implant allows the user to survive normally on such worlds, as it generates the requisite chemicals and antibodies to fend off those specific local hazards. It can be re-keyed to a different world with 2,000 credits worth of work at a medical clinic. The implantation itself inflicts 1 point of permanent System Strain.

Trailblazer: A specially-modified ATV explorer, a Trailblazer vehicle is equipped to cut through heavy vegetation and use advanced thermal processing techniques to turn local topsoils into a solid, effective road behind it. The trail created is only four meters wide, but it is sufficient to support wheeled traffic and can hold up to years of use with minimal maintenance. The Trailblazer can create bridges over small streams and creeks, but cannot take the road across rivers or canyons. The vehicle turns local vegetation into power for the road-building unit; if such vegetation is available, it can create ten kilometers of road a day through plains or low hills. Lacking it, another Type B power cell is consumed each day, along with the vehicle's usual power demand. Trailblazers all have pressurized, temperature-controlled atmospheres that can be regenerated for weeks on end.

Item	Cost	Enc
Biomass Generator	500	3
Floater Lamp	100	2
Mag-chain Cutter	250	2
Micro Comm Satellite	500	2
Native Antitoxin	50	1
Panic Shelter	200	1
Patch Putty	50	1
Perimeter Alarm	250	2
Poisonous Bait	50	1
Portable Ladder	100	1
Porterbot	2,000	-
Survival Guide	10	1
Tailored Adaptation Implant	2,000	-
Trailblazer	25,000	-
Void Skin Implant	15,000	-
Wayfinder Beacon	250	1
Wildlife Repellent Bomb	25	1

Void Skin Implant: This cybernetic augment pressure-seals the user when activated as an On Turn action, with a reinforced epidermis, oxygen cracker, limited radiation screen, worklight, and low-speed gas propulsion unit integrated into their body structure. When the unit is active the user can survive normal deep-space vacuum conditions and maneuver at normal movement rates in zero-g. The implant adds 1 point of permanent System Strain and can be activated for up to 12 hours before an equal amount of time must be spent with it deactivated.

Wayfinder Beacon: These compact radio beacons are designed to be powered by heavy Type B power cells but require no maintenance for years on end once placed. Their signals have a range of approximately 200 kilometers; if two beacon signals are available, a standard compad can use their transmissions to triangulate the user's current location even without the benefit of orbital GPS satellites. The beacon signals can be set to frequency-hop, making them largely invisible to civilian-grade radio monitoring. Each beacon maintains an audiovisual and weather log of its surroundings for up to several years.

Wildlife Repellent Bomb: The size of a large grenade, this bomb can be hurled up to 20 meters. When it explodes, it throws out a tremendously noxious cloud of gas with a 5-meter radius. Intelligent creatures in the cloud must make a Physical save or suffer a -2 penalty to hit rolls for the scene, while animals must immediately make a Morale check at a -2 penalty or flee the stench. Animals with no olfactory organs or that are driven by desperation are not affected by the cloud.